PRACTICE SHEET

1. What are coordinates of the point equidistant from the points (a, 0, 0), (0, a, 0), (0, 0, a) and (0, 0, 0)?
(a) $\left(\frac{a}{3},\frac{a}{3},\frac{a}{3}\right)$ (b) $\left(\frac{a}{2},\frac{a}{2},\frac{a}{2}\right)$
(c) (a, a, a) (d) (2a, 2a, 2a)

2. What is the length of the perpendicular from the origin to the plane $ax+by+\sqrt{2ab}z=1$?
(a) 1 / (ab) (b) 1 / (a + b)
(c) a + b (d) ab

3. How many arbitrary constants are there in the equation of a plane?
(a) 2 (b) 3
(c) 4 (d) Any finite number

4. If P, Q are (2, 5, –7), (–3, 2, 1) respectively, then what are the direction ratios of the line PQ?
(a) <10, 6, –16> (b) <5, 3, 8>
(c) <–5, –3, –8> (d) None of these

5. If O, P are the points (0, 0, 0), (2, 3, –1) respectively, then what is the equation to the plane through P at right angles to OP?
(a) 2x + 3y + z = 16 (b) 2x + 3y – z = 14
(c) 2x + 3y + z = 14 (d) 2x + 3y – z = 0

6. Which one of following is correct?
The three planes 2x + 3y – z –2 = 0, 3x + 3y + z – 4 = 0, x – y + 2z – 5 = 0 intersect
(a) at a point (b) at two points
(c) at three points (d) in a line

7. What is the centre of the sphere $ax^2 + by^2 + cz^2 - 6x = 0$ if the radius is 1 unit?
(a)(0, 0, 0)
(b)(1, 0, 0)
(c)(3, 0, 0)
(d)Cannot be determined as values of a, b, c are unknown

8. Under what condition do $\left\langle \frac{1}{\sqrt{2}},\frac{1}{2},K\right\rangle$ represent direction cosines of a line?
(a) $k=\frac{1}{2}$ (b) $k=-\frac{1}{2}$
(c) $k=\pm\frac{1}{2}$ (d) k can take any value

9. If x = a sec θ cos ϕ, y = b sec θ sin ϕ, z = c tan θ, then what is $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}$ equal to?
(a) 1 (b) 0
(c) – (d) $a^2 + b^2$

10. A line makes angles θ, ϕ and ψ with x,y,z axes respectively. Consider the following.
1. $\sin^2\theta + \sin^2\phi = \cos^2\psi$
2. $\cos^2\theta + \cos^2\phi = \sin^2\psi$
3. $\sin^2\theta + \cos2\theta = \cos^2\psi$
Which of the above is /are correct?
(a) 1 only (b) 2 only
(c) 3 only (d) 2 and 3

11. What is the ratio in which the line joining the points, (2, 4, 5) and (3, 5, –4) is internally divided by the xy – plane?
(a) 5 : 4 (b) 3 : 4
(c) 1 : 2 (d) 7 : 5

12. Under which one of the following conditions will the two planes x + y + z = 7 and αx + βy + γz = 3, be parallel (but not coincident)?
(a) α = β = γ = 1 only (b) $\alpha=\beta=\gamma=\frac{3}{7}$ only
(c) α = β = γ (d) None of the above

13. The straight line $\frac{x-3}{2}=\frac{y-4}{3}=\frac{z-5}{4}$ is parallel to which one of the following?
(a) 4x + 3y – 5z = 0 (b) 4x + 5y – 4z = 0
(c) 4x + 4y – 5z = 0 (d) 5x + 4y – 5z = 0

14. The equation *by* + *cz* + *d* = 0 represents a plane parallel to which one of the following?
(a) *x* – axis (b) *y* – axis
(c) z – axis (d) None of these

15. Which one of the following planes is normal to the plane 3*x* + *y* + z = 5?
(a) *x* + 2*y* + *z* = 6 (b) *x* – 2*y* + *z* = 6
(c) x + 2y – z = 6 (d) x – 2y – z = 6

16. If the radius of the sphere $x^2 + y^2 + z^2 - 6x - 8y + 10z + \lambda = 0$ is unity, what is the value of λ?
(a) 49 (b) 7
(c) – 49 (d) –7

17. Curve of intersection of two spheres is
(a) an ellipse (b) a circle
(c) a parabola (d) None of these

18. What is the number of planes passing through three non – collinear points?
(a) 3 (b) 2
(c) 1 (d) 0

19. What is the angle between the planes 2*x* – *y* + *z* = 6 and *x* + *y* + 2*z* = 3?
(a) π / 2 (b) π / 3
(c) π / 4 (d) π / 6

20. What is the value of *n* so that the angle between the lines having direction ratios (1, 1, 1) and (1, –1,*n*) is 60°?
(a) $\sqrt{3}$ (b) $\sqrt{6}$
(c) 3 (d) None

21. The two planes ax + by + cz + d = 0 and ax + by + cz + d_1 = 0, where d ≠d_1, have:
(a) One point only in common
(b) Three points in common
(c) Infinite points in common
(d) No point in common

22. What is the distance of the origin from the plane 2x + 6y – 3z + 7 = 0?
(a) 1 (b) 2
(c) 3 (d) 6

23. Under what condition do the planes bx – ay = n, cy – bz = *I*, az – cx = m interest in a line?
(a) a + b + c = 0 (b) a = b = c
(c) aI + bm + cn = 0 (d) I + m + n = 0

24. The planes px + 2y + 2z – 3 = 0 and 2x – y + z + 2 =0 intersect at an angle $\frac{\pi}{4}$. What is the value of p^2?
(a) 24 (b) 12

(c) 6 (d) 3

Directions (for next three) : The vertices of a cube are (0,0,0), (2,0,0), (0,0,2), (2,2,0), (2,0,2), (0,2,2), (2,2,2), respectively.

25. What is the angle between any two diagonals of the cube?

(a) $\cos^{-1}\left(\frac{1}{2}\right)$ (b) $\cos^{-1}\left(\frac{1}{3}\right)$

(c) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (d) $\cos^{-1}\left(\frac{2}{\sqrt{2}}\right)$

26. What is the angle between one of the edges of the cube and the diagonal of the cube intersecting the edge of the cube?

(a) $\cos^{-1}\left(\frac{1}{2}\right)$ (b) $\cos^{-1}\left(\frac{1}{3}\right)$

(c) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (d) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

27. What is the angle between the diagonal of one of the faces of the cube and the diagonal of the cube intersecting the diagonal of the face of the cube?

(a) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (b) $\cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(c) $\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$ (d) $\cos^{-1}\left(\frac{\sqrt{2}}{3}\right)$

28. If the line through the points A(k, 1, –1) and B(2k,0,2) is perpendicular to the line through the points B and C(2+2k, k,1), then what is the value of k?

(a) –1 (b) 1

(c) –3 (d) 3

29. Let O(0,0,0), P(3,4,5), Q (m, n, r) and R (1,1,1) be the vertices of a parallelogram taken in order. What is the value of m + n + r?

(a) 6 (b) 12

(c) 15 (d) 8

30. What is the equation of the plane through z-axis and parallel to the line $\frac{x-1}{\cos\theta}=\frac{y+2}{\sin\theta}=\frac{z-3}{0}$?

(a) $x\cot\theta + y = 0$ (b) $x\tan\theta - y = 0$

(c) $x + y\cos\theta = 0$ (d) $x - y\tan\theta = 0$

ANSWER KEYS																			
1.	b	**2.**	b	**3.**	c	**4.**	d	**5.**	b	**6.**	d	**7.**	d	**8.**	c	**9.**	a	**10.**	b
11.	a	**12.**	c	**13.**	c	**14.**	a	**15.**	d	**16.**	a	**17.**	b	**18.**	b	**19.**	b	**20.**	b
21.	d	**22.**	a	**23.**	c	**24.**	a	**25.**	b	**26.**	c	**27.**	c	**28.**	d	**29.**	c	**30.**	b

Solutions

Sol.1. (b)

Let the point A (x, y, z) is equidistant from the points B (a, 0, 0), C(0, a, 0), D(0,0,a) and E(0,0,0).

Hence,$(x-a)^2+y^2+z^2 = x^2+(y-a)^2+z^2$

$= x^2+y^2+(z-a)^2$

$= x^2+y^2+z^2$

$\Rightarrow (x-a)^2+y^2+z^2 = x^2+(y-a)^2+z^2$

$\Rightarrow x^2+a^2-2ax+y^2+z^2 = x^2+y^2+a^2-2ay+z^2$

$\Rightarrow -2ax = -2ay$

$\Rightarrow ax = ay \Rightarrow x = y$

Similarly, $ay = az \Rightarrow y = z$

$\Rightarrow x = y = z$

$\therefore (x-a)^2+x^2+x^2+x^2 = x^2+x^2$

$\Rightarrow x^2+z^2-2ax+x^2+x^2 = 3x^2$

$\Rightarrow a^2 = 2ax \Rightarrow x = a/2$

$\therefore$ point is $\left(\frac{a}{2},\frac{a}{2},\frac{a}{2}\right)$

Sol.2. (b)

Length of perpendicular from the origin to the plane $ax+by+\sqrt{2ab}z-1=0$ is

$=\left|\frac{0+0+0-1}{\sqrt{a^2+b^2+2ab}}\right|=\frac{1}{\sqrt{(a+b)^2}}=\frac{1}{(a+b)}$

Sol.3. (c)

Since, the general equation of a plane is ax + by + cz + d = 0. Where a, b, c show direction ratio and d is a parameter.

So, the number of arbitrary constants in equation of a plane = 4.

Sol.4. (d)

Since, coordinates of points P and Q are (2, 5, –7) and (–3, 2, 1) respectively.

Direction ratios of PQ are

<–3–2, 2–5, 1 + 7 > i.e., <–5, –3, 8>

Sol.5. (b)

Since, coordinates of points O and P are (0, 0, 0) and (2, 3, –1) respectively.

Direction ratios of OP are < 2, 3, –1>.

The plane is perpendicular to OP. So, its equation is $2x + 3y - z + d = 0$

Since, this pane passes through (2, 3, –1);

$2\times2+3\times3-1\times-1+d=0$

$\Rightarrow 4+9+1+d=0$

$\Rightarrow d = -14$

On putting the value of d in Eq. (i)

$2x+3y-z-14=0$

$\Rightarrow 2x+3y-z=14$

Which is required equation of plane.

Sol.6. (d)

Planes always intersect in a line

Sol.7. (d)

In the given equation, there are three unknown parameters and no equation has been given to evaluate those, hence centre of sphere cannot be determined.

Sol.8. (c)

For $\left(\frac{1}{\sqrt{2}},+\frac{1}{2},k\right)$, to represent direction cosines

$\left(\frac{1}{\sqrt{2}}\right)^2+\left(\frac{1}{2}\right)^2+k^2=1$

Or $1/2 + 1/4 + k^2 = 1$

$3/4 + k^2 = 1 \Rightarrow k = \pm 1/2$

Sol.9. (a)

As given

$x = a\sec\theta\cos\phi,\ y = b\sec\theta\sin\phi,\ z = c\tan\theta$

so, $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=\frac{a^2\sec^2\theta\cos^2\phi}{a^2}$

$+\frac{b^2\sec^2\theta\sin^2\phi}{b^2}-\frac{c^2\tan^2\theta}{c^2}$

$= \sec^2\theta(\cos^2\phi+\sin^2\phi)-\tan^2\theta = \sec^2\theta-\tan^2\theta = 1$

Sol.10. (b)

If a line makes angle θ, ϕ and ψ with x, y, z axes respectively, then

$\cos^2\theta+\cos^2\phi+\cos^2\psi = 1$

$\Rightarrow \cos^2\theta+\cos^2\phi = 1-\cos^2\psi \Rightarrow \cos^2\theta+\cos^2\phi = \sin^2\psi$

$\therefore$ Statement (2) is correct.

Sol.11. (a)

Let the line joining the points (2, 4, 5) and (3, 5, –4) is internally divided by the xy – plane in the ratio k : 1.

$\therefore$ For xy plane, z = 0

$\Rightarrow 0 = \frac{-k \times 4 + 5}{k+1} \Rightarrow 4k = 5 \Rightarrow k = \frac{5}{4}$

$k = \frac{5}{4}$ so, ratio is 5 : 4

Sol.12. (c)

Given equation of planes are: x+y+z=7

and $\alpha x + \beta y + \gamma z = 3$

For these planes to be parallel, coefficients of x, y and z should be same i.e. $\Rightarrow \alpha = \beta = \gamma$

Sol.13. (c)

A plane ax + by + cz = 0 is parallel to, a straight line having direction ratios a', b', c'.

If aa' + bb' + cc' = 0

In the given problem, dr_s of line is 2, 3, 4

We check the equations of plane in the given choices, one by one.

(a) $4 \times 2 + 3 \times 3 + (-5) \times 4 = 8 + 9 - 20 \neq 0$

(b) $4 \times 2 + 5 \times 3 + (-4) \times 4 = 8 + 15 - 16 \neq 0$

(c) $4 \times 2 + 4 \times 3 + (-5) \times 4 = 8 + 12 - 20 = 0$

Sol.14. (a)

Direction cosines of the normal to the given plane $by + cz + d = 0$ are 0, b, c

Direction consines of the x – axis are 1, 0, 0

Since, $0 \times 1 + b \times 0 + c \times 0 = 0$

Hence x – axis is perpendicular to normal to the given plane. Therefore x – axis is parallel to the given plane.

Sol.15. (d)

Direction cosines of the normal to the plane 3x + y + z = 5 are 3, 1, 1

Direction cosines of the normal to the plane

x – 2y – z = 6 are 1, –2, –1

Sum of the product of direction cosines

$= 3 \times 1 + 1 \times (-2) + 1 \times (-1) = 0$

Hence, normal's to the two planes are perpendicular to each other. Therefore two planes are also perpendicular.

Sol.16. (a)

Given sphere:

$x^2 + y^2 + z^2 - 6x - 8y + 10^z + \lambda = 0$

Its radius = 1

$\Rightarrow \sqrt{(-3)^2 + (-4)^2 + (5)^2 - \lambda} = 1$

$\Rightarrow 9 + 16 + 25 - \lambda = 1$

$\therefore \lambda = 49$

Sol.17. (b)

a circle (obviously)

Sol.18. (c)

We know that the number of planes passing through the non – collinear points is 1.

Sol.19. (b)

$a_1 x + b_1 y + c_1 z = d_1$ and

$a_2 x + b_2 y + c_2 z = d_2$

are two planes then angle between them is

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Let θ be the angle between given planes

Here, $a_1 = 2$, $b_1 = -1$, $c_1 = 1$

$a_2 = 1$, $b_2 = 1$, $c_2 = 2$

$$\therefore \cos\theta = \frac{2 \times 1 + 1 \times (-1) + 1 \times 2}{\sqrt{4+1+1}\sqrt{1+1+4}}$$

$= 3/6 = 1/2 = \cos \pi/3$

$\Rightarrow \theta = \pi/3$

Sol.20. (b)

If (l_1, m_1, n_1) and (l_2, m_2, n_2) are the direction ratios then angle between the lines is

$$\therefore \cos\theta = \frac{l_1l_2 + m_1m_2 + n_1n_2}{\sqrt{l_1^2 + m_1^2 + n_1^2}\sqrt{l_2^2 + m_2^2 + n_2^2}}$$

Here $l_1 = 1$, $m_1 = 1$, $n_1 = 1$ and $l_2 = 1$, $m_2 = -1$, $n_2 = n$ and $\theta = 60°$

$$\therefore \cos 60° = \frac{1 \times 1 + 1 \times (-1) + 1 \times n}{\sqrt{1^2 + 1^2 + 1^2}\sqrt{1^2 + 1^2 + n^2}}$$

$= 1/2 = \frac{n}{\sqrt{3}\sqrt{2+n^2}} \Rightarrow 3(2+n^2) = 4n^2$

$\Rightarrow n^2 = 6 \Rightarrow n = \pm\sqrt{6}$

Sol.21. (d)

These planes are parallel

Sol.22. (a)

$$D = \left|\frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}\right|$$

$$D = \left|\frac{0+0+0+7}{\sqrt{4+36+9}}\right| = 1$$

Sol.23. (c)

These planes intersect in a line al + bm + cn = 0

Sol.24. (a)

$$\cos\theta = \left|\frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}\right|$$

$$\cos\frac{\pi}{4} = \left|\frac{p(2) + 2(-1) + 2(1)}{\sqrt{p^2 + 4 + 4}\sqrt{4+1+1}}\right| = \frac{1}{\sqrt{2}}$$

by solving it $p^2 = 24$

Sol.25. (b)

Sol.26. (c)

Sol.27. (c)

Sol.28. (d)

D.R. of AB < k, −1, 3 >

D.R of BC < 2, k, −1 >

lines are perpendicular

so 2k −k −3 = 0

k = 3

Sol.29. (c)

Mid-point of OQ is mid-point of PR

Sol.30. (b)

Let equation of plane through z axis is

ax + by = 0

It is given that this plane is parallel to the line

$$\frac{x-1}{\cos\theta} = \frac{y+2}{\sin\theta} = \frac{z-3}{0}$$

Since the plane is parallel to line

$a\cos\theta + b\sin\theta = 0$

$a = -b\tan\theta$

so $-b\tan\theta x + by = 0$

$x\tan\theta - y = 0$

NDA PYQ

1. What is the angle between the lines whose direction cosines are proportional to (2,3,4) and (1,–2,1), respectively?
(a) 90° (b) 60°
(c) 45° (d) 30°
[NDA (I)-2011]

2. What is the radius of the sphere $x^2 + y^2 + z^2 - x - z = 0$?
(a) $\sqrt{\frac{3}{4}}$ (b) $\sqrt{\frac{1}{2}}$
(c) $\sqrt{\frac{3}{2}}$ (d) $\frac{1}{3}$
[NDA (I)-2011]

3. Consider the following relations among angles α, β and γ made by a vector with the coordinate axes.
I. $\cos2\alpha + \cos2\beta + \cos2\gamma = -1$
II. $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$
Which of the above statement is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (I)-2011]

4. What is the acute angle between the planes $x + y + 2z = 3$ and $-2x + y - z = 11$?
(a) π/5 (b) π/4
(c) π/6 (d) π/3
[NDA (I) - 2011]

5. Which of the following point lie on the plane $2x + 3y - 6z = 21$?
(a) (3,2,2) (b) (3,7,1)
(c) (1,2,3) (d) (2,1,−1)
[NDA (I)-2011]

6. What is the locus of points of intersection of a sphere and a plane?
(a) Circle (b) Ellipse
(c) Parabola (d) Hyperbola
[NDA (II) - 2011]

7. What is angle between two planes $2x-y+z=4$ and $x+y+2z=6$?
(a) π/2 (b) π/3
(c) π/4 (d) π/6
[NDA (II) - 2011]

8. What is the equation of the plane passing through the point (1,−1,−1) and perpendicular to each of the planes $x-2y-8z=0$ and $2x+5y-z=0$?
(a) $7x-3y+2z=14$ (b) $2x+5y-3z=12$
(c) $x-7y+3z=4$ (d) $14x-5y+3z=16$
[NDA (II)-2011]

9. The equation to sphere passing through origin and the points (−1,0,0),(0, −2,0) and (0,0,−3)is $x^2 + y^2 + z^2 + f(x,y,z) = 0$. what is f(x,y,z) =
(a) $-x-2y-3z$ (b) $x + 2y + 3z$
(c) $x + 2y + 3z - 1$ (d) $x + 2y + 3z + 1$
[NDA (II)-2011]

10. What are the direction ratios of normal to the plane $2x - y + 2z + 1 = 0$?
(a) <2, 1, 2> (b) $\left\langle 1, -\frac{1}{2}, 1 \right\rangle$
(c) $\langle 1, -2, 1 \rangle$ (d) $\langle 1, 1, 2 \rangle$
[NDA (I) - 2012]

11. What is the cosine of angle between the planes $x + y + z + 1 = 0$ and $2x - 2y + 2z + 1 = 0$?
(a) 1/2 (b) 1/3
(c) 2/3 (d) 1/4
[NDA (I) - 2012]

12. What is the diameter of the sphere $x^2 + y^2 + z^2 - 4x + 6y - 8z - 7 = 0$?
(a) 4 units (b) 5 units
(c) 6 units (d) 12 units
[NDA (I) - 2012]

13. What is the sum of the squares of direction cosines of the line joining the points (1, 2, –3) and (–2, 3, 1)?
(a) 0 (b) 1
(c) 3 (d) $\frac{2}{\sqrt{26}}$
[NDA (I) - 2012]

14. If a line makes the angles α , β , γ with the axes, then what is the value of $1 + \cos2\alpha + \cos2\beta + \cos2\gamma$ equal to
(a) −1 (b) 0
(c) 1 (d) 2
[NDA (I) - 2012]

15. What are the direction ratios of the line of intersection of the planes $x=3z+4$ and $y=2z-3$?
(a) $\langle 1,2,3 \rangle$ (b) $\langle 2,1,3 \rangle$
(c) $\langle 3,2,1 \rangle$ (d) $\langle 1,3,2 \rangle$
[NDA (II) - 2012]

16. What is the equation of the straight line passing through (a, b, c) and parallel to Z-axis?
(a) $\frac{x-a}{1} = \frac{y-b}{0} = \frac{z-c}{0}$ (b) $\frac{x-a}{0} = \frac{y-b}{0} = \frac{z-c}{1}$
(c) $\frac{x-a}{0} = \frac{y-b}{1} = \frac{z-c}{0}$ (d) $\frac{x-a}{0} = \frac{y-b}{1} = \frac{z-c}{1}$
[NDA (II) - 2012]

17. What is the equation to the plane through (1, 2, 3) parallel to $3x + 4y - 5z = 0$?
(a) $3x + 4y + 5z + 4 = 0$
(b) $3x + 4y - 5z + 14 = 0$
(c) $3x + 4y - 5z + 4 = 0$
(d) $3x + 4y - 5z - 4 = 0$
[NDA - (II) 2012]

18. What is the distance of the point (1, 2, 0) from YZ-plane is
(a) 1 unit (b) 2 units
(c) 3 units (d) 4 units
[NDA (II) - 2012]

19. If a line OP of length r (where, O is the origin) makes an angle α with X-axis and lies in the XZ-plane, then what are the coordinates of P?
(a) (r cos α, 0, r sin α) (b) (0, 0, r sin α)
(c) (r cos α, 0, 0) (d) (0, 0, r cos α)
[NDA (II) - 2012]

20. If the distance between the points (7, 1, –3) and (4, 5, λ) is 13 units, then what is one of the values of λ?
(a) 20 (b) 10
(c) 9 (d) 8
[NDA (II) - 2012]

21. What is the angle between the lines $\frac{x-2}{1} = \frac{y+1}{-2} = \frac{z+2}{1}$ and $\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2}$?

(a) $\pi/2$ (b) $\pi/3$
(c) $\pi/6$ (d) None of these

[NDA (II) - 2012]

22. What are the direction cosines of a line which is equally inclined to the positive directions of the axes?

(a) $\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$ (b) $\left\langle -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$

(c) $\left\langle -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$ (d) $\left\langle \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right\rangle$

[NDA (II) - 2012]

23. What is the angle between the planes $2x - y - 2z + 1 = 0$ and $3x - 4y + 5z - 3 = 0$?

(a) $\pi/6$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$

[NDA (I) - 2013]

24. If the straight line $\frac{x-x_0}{\ell} = \frac{y-y_0}{\imath} = \frac{z-z_0}{n}$ is parallel to the plane $ax + by + cz + d = 0$, then which one of the following is correct?

(a) $l + m + n = 0$ (b) $a + b + c = 0$

(c) $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} = 0$ (d) $al + bm + cn = 0$

[NDA (I) - 2013]

25. What is the distance between the planes $x - 2y + z - 1 = 0$ and $-3x + 6y - 3z + 2 = 0$?

(a) 3 units (b) 1 unit
(c) 0 (d) None of these

[NDA (I) - 2013]

26. What should be the value of k for which the equation $3x^2 + 3y^2 + (k + 1)z^2 + x - y + z = 0$, represents the sphere?

(a) 3 (b) 2
(c) 1 (d) –1

[NDA (I) - 2013]

27. If a line makes 30° with the positive direction of X-axis, $\angle\beta$ with the positive direction of Y-axis and $\angle\gamma$ with the positive direction of Z-axis, then what is $\cos^2\beta + \cos^2\gamma$ equal to?

(a) 1/4 (b) 1 /2
(c) 3/4 (d) 1

[NDA (I) - 2013]

28. The sum of the direction cosines of Z-axis is

(a) 0 (b) 1/3
(c) 1 (d) 3

[NDA (I) - 2013]

29. If θ is the acute angle between the diagonals of a cube, then which one of the following is correct?

(a) $\theta = 30°$ (b) $\theta = 45°$
(c) $2\cos\theta = 1$ (d) $3\cos\theta = 1$

[NDA (II) - 2013]

30. What is the equation of the sphere with unit radius having center at the origin?

(a) $x^2 + y^2 + z^2 = 0$ (b) $x^2 + y^2 + z^2 = 1$
(c) $x^2 + y^2 + z^2 = 0$ (d) $x^2 + y^2 + z^2 = 3$

[NDA (II) - 2013]

31. What is the sum of the squares of direction cosines of X-axis?

(a) 0 (b)1/3
(c) 1 (d) 3

[NDA (II) - 2013]

32. What is the distance of the line $2x + y + 2z = 3$ from the origin?

(a) 1 unit (b) 1.5 units
(c) 2 units (d) 2.5 units

[NDA (II) - 2013]

33. If the projection of a straight line segment on the coordinate axes are 2,3,6 then the length of the segment is

(a) 5 units (b) 7 units
(c) 11 units (d) 49 units

[NDA (II) - 2013]

Directions (for next two): Consider the spheres $x^2 + y^2 + z^2 - 4y+3 = 0$ and $x^2 + y^2 + z^2 + 2x + 4z - 4 = 0$

34. What is the distance between the centers of the two spheres?

(a) 5 units (b) 4 units
(c) 3 unit (d) 2 units

[NDA (I) - 2014]

35. Consider the following statements

I. The two spheres intersect each other.

II.The radius of first sphere is less than that of second sphere.

Which one of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) - 2014]

36. If a line passes through the points (6, –7, –1) and (2, –3, 1), then what are the direction ratios of the line?

(a) $\langle 4, -4, 2\rangle$ (b) $\langle 4, 4, 2\rangle$

(c) $\langle -4, 4, 2\rangle$ (d) $\langle 2, 1, 1\rangle$

[NDA (I) - 2014]

Directions (for next three): A straight line passes through (1, –2, 3) and perpendicular to the plane $2x + 3y - z = 7$.

37. What are the direction ratios of normal to plane?

(a) <2, 3, –1> (b) <2, 3, 1>
(c) <–1, 2, 3> (d) None of these

[NDA (I) - 2014]

38. Where does the line meet the plane?

(a) (2, 3, –1) (b) (1, 2, 3)
(c) (2, 1, 3) (d) (3, 1, 2)

[NDA (I) - 2014]

39. What is the image of the point (1, –2, 3) in the plane?

(a) (2, –1, 5) (b) (–1, 2, –2)
(c) (5, 4, 1) (d) None of these

[NDA (I) - 2014]

For the next two (02) items that follow:

Let a vector $\vec{r}$ make angles 60°, 30° with x and y-axis respectively.

40. What angle does $\vec{r}$ make with z-axis?

(a) 30° (b) 60°
(c) 90° (d) 120°

[NDA-2014(1)]

41. What are the direction cosines of $\vec{r}$?

(a) $\left\langle \frac{1}{2}, \frac{\sqrt{3}}{2}, 0 \right\rangle$ (b) $\left\langle \frac{1}{2}, -\frac{\sqrt{3}}{2}, 0 \right\rangle$

(c) $\left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right\rangle$ (d) $\left\langle -\frac{1}{2}, \frac{\sqrt{3}}{2}, 0 \right\rangle$

[NDA-2014(1)]

Directions (for next two): Read the following information carefully and answer the question given below.

Consider the plane passing through the points A (2, 2, 1), B(3, 4, 2) and C(7, 0, 6).

42. Which one of the following points lies on the plane?

(a) (1, 0, 0) (b) (1, 0, 1)
(c) (0, 0, 1) (d) None of these

[NDA (II) - 2014]

43. What are the direction ratios of the normal to the plane?
(a) <1, 0, 1> (b) < 0, 1, 0>
(c) <1, 0, –1> (d) None of these
[NDA (II) - 2014]

Directions (for next three): Read the following information carefully and answer the question given below. Consider a sphere passing through the origin and the points (2, 1, –1), (1, 5, –4), (–2, 4, –6).

44. What is the radius of the sphere?
(a) $\sqrt{12}$ (b) $\sqrt{14}$
(c) 12 (d) 14
[NDA (II) - 2014]

45. What is the center of the sphere?
(a) (–1, 2, –3) (b) (1, –2, 3)
(c) (1, 2, –3) (d) (–1, –2, –3)
[NDA (II) - 2014]

46. Consider the following statements
I. The sphere passes through the point (0, 4, 0).
II. The point (1, 1, 1) is at a distance of 5 unit from the center of the sphere.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
[NDA (II) - 2014]

Directions (for next two): The line joining the points (2,1,3) and (4,–2,5) cuts the plane 2x+y–z=3

47. Where does the line cut the plane
(a) (0,–4, –1) (b) (0,–4, 1)
(c) (1,4, 0) (d) (0, 4 , 1)
[NDA (II) - 2014]

48. What is the ratio in which the plane divided the line?
(a) 1:1 (b) 2:3
(c) 3:4 (d) none of these
[NDA (II) - 2014]

Directions (for next two): The projections of a directed line segment on the coordinate axes are 12, 4, 3 respectively.

49. What is the length of line segment?
(a) 19 units (b) 17 units
(c) 15 units (d) 13 units
[NDA (I) - 2015]

50. What are the direction cosines of the line segment?
(a) $\left(\pm\frac{12}{13},\pm\frac{4}{13},\frac{3}{13}\right)$ (b) $\left(\frac{12}{13},-\frac{4}{31},\frac{3}{13}\right)$
(c) $\left(\frac{12}{13},-\frac{4}{13},-\frac{3}{13}\right)$ (d) $\left(-\frac{12}{13},-\frac{4}{13},\frac{3}{13}\right)$
[NDA (I) - 2015]

Direction (for next two): From the point P(3, –1, 11), a perpendicular is drawn on the line L given by the equation $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$. Let Q be the foot of the perpendicular.

51. What are the direction ratios of the line segment PQ?
(a) (1, 6, 4) (b) (–1, 6, –4)
(c) (–1, –6, 4) (d) (2, –6, 4)
[NDA (I) - 2015]

52. What is the length of the line segment PQ?
(a) $\sqrt{47}$ unit (b) 7 unit
(c) $\sqrt{53}$ unit (d) 8 unit
[NDA (I) - 2015]

Directions (for next two): A triangular plane ABC with centroid (1, 2, 3) cuts the coordinates axes at A, B, C, respectively.

53. What are the intercepts made by the plane ABC on the axes?
(a) 3, 6, 9 (b) 1, 2, 3
(c) 1, 4, 9 (d) 2, 4, 6
[NDA (I) - 2015]

54. What is the equation of the plane ABC?
(a) x + 2y + 3z = 1 (b) 3x + 2y + z = 3
(c) 2x + 3y + 6z = 18 (d) 6x + 3y + 2z = 18
[NDA (I) - 2015]

Directions (for next two): A point P(1, 2, 3) is one vertex of a cuboids formed by the coordinates planes and the planes passing through P and parallel to the coordinates planes.

55. What is the length of one of the diagonals of the cuboids?
(a) $\sqrt{10}$ unit (b) $\sqrt{14}$ unit
(c) 4 unit (d) 5 unit
[NDA (I) - 2015]

56. What is the equation of plane passing through P(1, 2, 3) and parallel to XY-plane?
(a) x + y = 3 (b) x – y = –1
(c) z = 3 (d) 2y + 3zx = 14
[NDA (I) - 2015]

57. The lines 2x = 3y = –z and 6x = –y = –4z?
(a) Are perpendicular
(b) Are parallel
(c) Intersect at an angle 45°
(d) Intersect at an angle 60°
[NDA (II) - 2015]

58. The radius of the sphere
$3x^2 + 3y^2 + 3z^2 - 8x + 4y + 8z - 15 = 0$ is?
(a) 2 (b) 3
(c) 4 (d) 5
[NDA (II) - 2015]

59. The direction ratios of the line perpendicular to the lines with direction ratio <1, –2, –2> and <0, 2, 1> are
(a) <2, –1, 2> (b) <–2, 1, 2>
(c) <2, 1, –2> (d) <–2, –1, –2>
[NDA (II) - 2015]

60. What are the coordinates of the foot of the perpendicular drawn from the point (3, 5, 4) on the plane z = 0?
(a) (0, 5, 4) (b) (3, 5, 0)
(c) (3, 0, 4) (d) (0, 0, 4)
[NDA (II) - 2015]

61. The length of the intercepts on the coordinates axes made by the plane 5x + 2y + z – 13 = 0 are
(a) (5, 2, 1) units (b) $\left(\frac{13}{5},\frac{13}{2},13\right)$ units
(c) $\left(\frac{5}{13},\frac{2}{13},\frac{1}{13}\right)$ units (d) (1, 2, 5) units
[NDA (II) - 2015]

Directions (for next three): A plane P passes through the line of intersection of the planes 2x – y + 3z = 2, x + y –z = 1 and the point (1, 0, 1).

62. What are the direction ratios of the line of intersection of the given planes?
(a) < 2, –5, –3 > (b) < 1, –5, –3 >
(c) < 2, 5, 3 > (d) < 1, 3, 5 >
[NDA (I) - 2016]

63. What is the equation of the plane P?
(a) 2x + 5y – 2 = 0 (b) 5x + 2y – 5 = 0
(c) x + z –2 = 0 (d) 2x–y–2z = 0
[NDA (I) - 2016]

64. If the plane P touches the sphere $x^2 + y^2 + z^2 = r^2$, then what is r equal to?

(a) $\frac{2}{\sqrt{29}}$ units (b) $\frac{4}{\sqrt{29}}$ units

(c) $\frac{5}{\sqrt{29}}$ units (d) 1 unit

[NDA (I) - 2016]

Directions (for next two): Consider the following for the next two items that follow:

Let Q be the image of the point P (–2, 1, –5) in the plane $3x - 2y + 2z + 1 = 0$.

65. Consider the following:

1.The coordinates of Q are (4, –3, –1).

2.PQ is of length more than 8 units.

3.The point (1, –1, –3) is the mid-point of the line segment PQ and lies on the given plane.

Which of the above statements are correct?

(a) Both 1 and 2 (b) Both 2 and 3

(c) Both 1 and 3 (d) 1, 2 and 3

[NDA (II) - 2016]

66. Consider the following :

1.The direction ratios of the line segment PQ are <3, –2, 2 >.

2.The sun of the squares of direction cosines of the line segment PQ is unity.

Which of the above statements is / are correct?

(a) Only 1 (b) Only 2

(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (II) - 2016]

Directions (for next two): Consider the following for the next two items that follow:

A line L passes through the point P (5, –6, 7) and is parallel to the planes $x + y + z = 1$ and $2x - y - 2z = 3$.

67. What are the direction ratios of the line of intersection of the given planes?

(a) <1, 4, 3> (b) <–1, –4, 3 >

(c) <1, –4, 3 > (d) <1, –4, –3 >

[NDA (II) - 2016]

68. What is the equation of the line L?

(a) $\frac{x-5}{-1} = \frac{y+6}{4} = \frac{z-7}{-3}$ (b) $\frac{x+5}{-1} = \frac{y-6}{4} = \frac{z+7}{-3}$

(c) $\frac{x-5}{-1} = \frac{y+6}{-4} = \frac{z-7}{3}$ (d) $\frac{x-5}{-1} = \frac{y+6}{-4} = \frac{z-7}{-3}$

[NDA (II) - 2016]

69. A straight line with direction cosines (0, 1, 0) is:

(a) Parallel to x-axis

(b) Parallel to y-axis

(c) Parallel to z-axis

(d) Equally inclined to all the axes

[NDA (I) - 2017]

70. (0,0,0), (a, b, 0), (0,b,0) and (0,0,c) are four distinct points. What are the coordinates of the point which is equidistant from the four points?

(a) $\left(\frac{a+b+c}{3}, \frac{a+b+c}{3}, \frac{a+b+c}{3}\right)$

(b) (a, b, c)

(c) $\left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$

(d) $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$

[NDA (I) - 2017]

71. The line passing through the points (1, 2, –1) and (3, –1, 2) meets the yz plane at which one of the following points?

(a) $\left(0, -\frac{7}{2}, \frac{5}{2}\right)$ (b) $\left(0, \frac{7}{2}, \frac{1}{7}\right)$

(c) $\left(0, -\frac{7}{2}, -\frac{5}{2}\right)$ (d) $\left(0, \frac{7}{2}, -\frac{5}{2}\right)$

[NDA (I) - 2017]

72. Under which one of the following conditions are the lines $x = ay + b$; $z = cy + d$ and $x = ey + f$; $z = gy + h$ perpendicular?

(a) $ae + cg - 1 = 0$ (b) $ae + bf - 1 = 0$

(c) $ae + cg + 1 = 0$ (d) $ag + ce + 1 = 0$

[NDA (I) - 2017]

73. The point P (3,2,4), Q(4,5,2), R(5,8,0) and S(2, –1,6) are:

(a) Vertices of a rhombus which is not a square

(b) Non coplanar

(c) Collinear

(d) Coplanar but not collinear

[NDA (I) - 2017]

74. The length of the normal from origin to the plane $x + 2y - 2z = 9$ is equal to:

(a) 2units (b) 3units

(c) 4units (d) 5units

[NDA (II) - 2017]

75. The point of intersection of the line joining the points (–3, 4, –8) and (5, –6, 4) with the XY plane is:

(a) $\left(\frac{7}{3}, -\frac{8}{3}, 0\right)$ (b) $\left(-\frac{7}{3}, -\frac{8}{3}, 0\right)$

(c) $\left(-\frac{7}{3}, \frac{8}{3}, 0\right)$ (d) $\left(\frac{7}{3}, \frac{8}{3}, 0\right)$

[NDA (II) - 2017]

76. The equation of the plane passing through the line of intersection of the planes $x + y + z = 1$, $2x + 3y + 4z = 7$, and perpendicular to the plane $x - 5y + 3z = 5$ is given by:

(a) $x + 2y + 3z - 6 = 0$ (b) $x + 2y + 3z + 6 = 0$

(c) $3x + 4y + 5z - 8 = 0$ (d) $3x + 4y + 5z + 8 = 0$

[NDA (II) - 2017]

77. If the angle between the lines whose direction ratios are (2, –1, 2) and (x, 3, 5) is $\frac{\pi}{4}$, then the smaller value of x is:

(a) 52 (b) 4

(c) 2 (d) 1

[NDA (II) - 2017]

78. A variable plane passes through a fixed point (a, b, c) and cuts the axes in A, B and C respectively. The locus of the centre of the sphere OABC, O being the origin, is:

(a) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ (b) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$

(c) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$ (d) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 2$

[NDA (II) - 2017]

79. What is the equation to the sphere whose centre is at (–2, 3, 4) and radius is 6 units.

(a) $x^2 + y^2 + z^2 + 4x - 6y - 8z = 7$

(b) $x^2 + y^2 + z^2 + 6x - 4y - 8z = 7$

(c) $x^2 + y^2 + z^2 + 4x - 6y - 8z = 4$

(d) $x^2 + y^2 + z^2 + 4x + 6y + 8z = 4$

[NDA (I)-2018]

80. The coordinates of the vertices P, Q and R of a triangle PQR are (1, –1,1), (3,–2,2) and (0,2,6) respectively. If $\angle RQP = \theta$, then what is $\angle PRQ$ equal to?
(a) $30° + \theta$ (b) $45° – \theta$
(c) $60° – \theta$ (d) $90° – \theta$
[NDA (I)-2018]

81. A sphere of constant radius r through the origin intersects the coordinate axes in A, B and C. What is the locus of the centroid of the triangle ABC?
(a) $x^2 + y^2 + z^2 = r^2$ (b) $x^2 + y^2 + z^2 = 4r^2$
(c) $9(x^2 + y^2 + z^2) = 4r^2$ (d) $3(x^2 + y^2 + z^2)=2r^2$
[NDA (I)-2018]

82. What is the equation of the plane passing through the points (–2,6,–6), (–3,10,–9) and (–5,0,–6)?
(a) $2x – y – 2z = 2$ (b) $2x + y + 3z = 0$
(c) $x + y + z = 6$ (d) $x – y – z = 3$
[NDA (I)-2018]

83. Let the coordinates of the points A, B, C be (1,8,4), (0,–11,4) and (2, –3,1) respectively. What are the coordinates of the point D which is the foot of the perpendicular from A on BC?
(a) (3,4,–2) (b) (4,–2,5)
(c) (4,5, –2) (d) (2,4,5)
[NDA (I)-2018]

84. Consider the following statement:
1. The angle between the planes
$2x – y + z = 1$ and $x + y + 2z = 3$ is $\frac{\pi}{3}$
2. The distance between the planes
$6x – 3y + 6z + 2 = 0$ and $2x – y + 2z + 4=0$ is $\frac{10}{9}$
Which of the above statement is/are correct.
[NDA (II)-2018]
(a) 1 Only (b) 2 Only
(c) Both and 2 (d) Neither 1 nor 2

85. The line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is given by
(a) $x + y + z = 6,\ x + 2y – 3z = – 4$
(b) $x + 2y – 2z = – 1,\ 4x + 4y – 5z – 3 = 0$
(c) $3x + 2y – 3z = 0,\ 3x – 6y + 3z = – 2$
(d) $3x + 2y – 3z = – 2,\ 3x – 6y + 3z = 0$
[NDA (II)-2018]

86. Coordinates of the points O, P, Q and R are respectively (0,0,0), (4,6,2m), (2,0, 2n) and (2, 4, 6). Let L, M, N and K be points on the sides OR, OP, PQ and QR respectively such that LMNK is a parallelogram whose two adjacent sides side LK are each of length √2 What are the values of m and n respectively?
(a) 6, 2
(b) 1, 3
(c) 3, 1
(d) None of the above
[NDA (II)-2018]

87. What is the distance of the point (2,3,4) from the plane $3x – 6y + 2z + 11 = $?
(a) 1unit (b) 2 unit
(c) 3unit (d) 4 units
[NDA (II)-2018]

88. The centroid of the triangle with vertices A(2, –3,3), B(5, –3,–4) and C(2, –3, –2) is the point.
(a) (–3, 3, –1) (b) (3, –3, –1)
(c) (3, 1, –3) (d) (–3, –1, –3)

[NDA-2019(1)]

89. What are the direction cosines of z axis?
(a) <1, 1, 1> (b) <1, 0, 0>
(c) <0, 1, 1> (d) <0, 0, 1>
[NDA (I)-2019]

90. The distance between the parallel planes $4x−2y+4z+9=0$ and $8x−4y+8z+21=0$ is
(a) 1/4 (b) 1/2
(c) 3/2 (d) 7/4
[NDA (I)-2019]

91. The equation of plane passing through the intersection of the planes $2x + y + 2z = 9$, $4x−5y−4z = 1$ and the point (3,2,1) is
(a) $10x −2y + 2z = 28$ (b) $10x+2y+ 2z = 28$
(c) $10x −2y − 2z = 28$ (d) $10x−2y− 2z = 24$
[NDA (I)-2019]

92. What is the radius of the sphere $x^2 + y^2 + z^2 −6x + 8y −10z +1 = 0$?
(a) 5 (b) 2
(c) 7 (d) 3
[NDA (I)-2019]

93. A straight line passes through the point (1, 1, 1) makes an angle 60^0 with the positive direction of z-axis, and the cosines of the angles made by it with the positive direction of y-axis and x-axis are in the ratio √3:1. What is the acute angle between the two possible positions of the line?
(a) 90^0 (b) 60^0
(c) 45^0 (d) 30^0
[NDA (II)-2019]

94. A point on a line has coordinates $(p + 1, p − 3, \sqrt{2}\, p)$ where p is any number. What are direction cosines of the line?
(a) $\frac{1}{2},\frac{1}{2},\frac{1}{\sqrt{2}}$
(b) $\frac{1}{\sqrt{2}},\frac{1}{2},\frac{1}{2}$
(c) $\frac{1}{\sqrt{2}},\frac{1}{2},-\frac{1}{2}$
(d)Cannot be determined due to insufficient data
[NDA (II)-2019]

95. A point on the line $\frac{x-1}{1}=\frac{y-3}{2}=\frac{z+2}{7}$ has coordinates
(a) (3, 5, 4) (b) (2, 5, 5)
(c) (−1,−1,5) (d) (2,−1,0)
[NDA (II)-2019]

96. If the line $\frac{x-4}{1}=\frac{y-2}{1}=\frac{z-k}{2}$ lies on the plane $2x − 4y + z = 7$, then what is the value of k?
(a) 2 (b) 3
(c) 5 (d) 7
[NDA (II)-2019]

97. If the points (x,y,−3), (2,0,−1) and (4,2,3) lie on a straight line, then what is the value of x and y respectively ?
(a) 1, −1 (b) −1, 1
(c) 0, 2 (d) 3, 4
[NDA (II)-2019]

98. What is the length of the diameter of the sphere whose centre is at (1,−2,3) and which touches the plane $6x−3y + 2z − 4 = 0$?
(a) 1 unit (b) 2 units
(c) 3 units (d) 4 units
[NDA 2020]

OFFICIUM SUPREMUM

99. What is the perpendicular distance from the point (2,3,4) to the line $\frac{x-0}{1}=\frac{y-0}{0}=\frac{z-0}{0}$?
(a) 6 units (b) 5 units
(c) 3 units (d) 2 units
[NDA 2020]

100. If a line has direction ratios < a + b ,b + c, c + a >, then what is the sum of squares of its direction cosines ?
(a) $(a+b+c)^2$ (b) 2(a + b + c)
(c) 3 (d) 1
[NDA 2020]

101. Into how many compartments do the coordinate's planes divide the space?
(a) 2 (b) 4
(c) 8 (d) 16
[NDA 2020]

102. What is the equation of the plane which cuts an intercept 5 units on the z-axis and is parallel to xy-plane?
(a) x + y =5 (b) z = 5
(c) z = 0 (d) x + y + z = 5
[NDA 2020]

103. What is the angle between the two lines having direction ratio {6, 3, 6} and {3, 3, 0}?
(a) $\pi/6$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$
[NDA (I) 2021]

104. If l, m, n are the direction cosines of the line x–1 = 2(y + 3) = 1 – z, then what is $l^4+m^4+n^4$ equal to?
(a) 1 (b) $\frac{11}{27}$
(c) $\frac{13}{27}$ (d) 4
[NDA (I) 2021]

105. What is the projection of the line segment joining A(1, 7, –5) and B(–3, 4, –2) on y-axis?
(a) 5 (b) 4
(c) 3 (d) 2
[NDA (I) 2021]

106. What is the number of possible values of k for which the line joining the points (k, 1, 3) and (1, –2, k + 1) also passes through the point (15, 2 –4)?
(a) Zero (b) One
(c) Two (d) Infinite
[NDA (I) 2021]

107. The foot of the perpendicular drawn from the origin to the plane x + y + z = 3 is:
(a) (0, 1, 2) (b) (0, 0, 3)
(c) (1, 1, 1) (d) (–1, 1, 3)
[NDA (I) 2021]

108. The locus of a point P(x, y, z) which moves in such a way that z = 7 is a:
(a) line parallel to x-axis
(b) line parallel to y-axis
(c) line parallel to z-axis
(d) line parallel to xy-plane
[NDA (II) 2021]

109. Consider the following statements:
1.A line in space can have infinitely many direction ratios
2.It is possible for certain line that the sum of the squares of direction cosines can be equal to sum of its direction cosines.
Which of the above statements is/are correct?
(a)1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
[NDA (II) 2021]

110. The xy-plane divides the line segment joining the points (–1, 3, 4) and (2,–5, 6)
(a) internally in the ratio 2 : 3
(b) internally in the ratio 3 : 2
(c) externally in the ratio 2 : 3
(d) externally in the ratio 2 : 1
[NDA (II) 2021]

111. The number of spheres of radius r touching the coordinate axes is:
(a) 4 (b) 6
(c) 8 (d) infinite
[NDA (II) 2021]

112. ABCDEFGH is a cuboid with base ABCD. Let A(0,0,0), B(12,0,0), C(12,6,0) and G(12,6,4) be the vertices. If α is the angle between AB and AG; β is the angle between AC and AG, then what is the value of $\cos2\alpha+\cos2\beta$?
(a) $\frac{40}{49}$ (b) $\frac{64}{49}$
(c) $\frac{120}{49}$ (d) $\frac{160}{49}$
[NDA (II) 2021]

Consider the following for the next three (03) items that follow:
The plane 6x + ky + 3z – 12 = 0 where k ≠ 0 meets the coordinate axes at A, B and C respectively. The equation of the sphere passing through the origin and A, B, C is $x^2+y^2+z^2-2x-3y-4x=0$.

113. What is the value of k?
(a) 3 (b) 4
(c) 6 (d) 12
[NDA (I) 2022]

114. If P is the perpendicular distance from the centre of the sphere to the plane, then which one of the following is correct?
(a) 0 < p < 0.5 (b) 0.5 < p < 1
(c) 1 < p < 1.5 (d) p > 1.5
[NDA (I) 2022]

115. What is the equation of the line through the origin and the centre of the sphere?
(a) x = y = z (b) 2x = 3y = 4z
(c) 6x = 3y = 4z (d) 6x = 4y = 3z
[NDA (I) 2022]

Consider the following for the next two (02) items that follow:
Let the plane $\frac{2x}{k}+\frac{2y}{3}+\frac{z}{3}=2$ pass through the point (2,3, –6).

116. What are the direction ratios of a normal to the plane?
(a) <3, 2, 1> (b) <2, 3, 6>
(c) <6, 3, 2> (d) <1, 2, 3>
[NDA (I) 2022]

117. If p, q and r are the intercepts made by the plane on the coordinate axes respectively, then what is (p + q + r) equal to?
(a) 10 (b) 11
(c) 12 (d) 13
[NDA (I) 2022]

118. Consider the points A(2, 4, 6), B(–2, –4,–2), C(4, 6, 4) and D(8, 14, 12). Which of the following statements is/are correct?
1.The points are the vertices of a rectangle ABCD
2.The mid-point of AC is the same as that of BD.

Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) neither 1 nor 2
[NDA 2022 (II)]

119. Consider the equation of a sphere $x^2 + y^2 + z^2 - 4x - 6y - 8z - 16 = 0$.
Which of the following statements is/are correct?
1.z-axis is tangent to the sphere
2.The centre of the sphere lies on the plane $x + y + z - 9 = 0$
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2
[NDA 2022 (II)]

120. A plane cuts intercepts 2, 2,1 on the coordinate axes. What are the direction cosines of the normal to the plane?
(a) <2/3, 2/3, 1/3> (b) <1/3, 2/3, 2/3>
(c) $<\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}>$ (d) $<\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}>$
[NDA 2022 (II)]

121. Consider the following statements:
1.The direction ratios of y-axis can be <0, 4, 0>
2.The direction ratio of a line perpendicular to z-axis can be <5,6,0>
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2
[NDA 2022 (II)]

Consider the following for the next two (02) items that follows:
Let A(1,–1, 2) and B(2, 1, –1) be the end points of the diameter of the sphere.
$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz - 1 = 0$

122. What is u + v + w equal to?
(a) –2 (b) –1
(c) 1 (d) 2
[NDA – 2023 (1)]

123. If P (x, y, z) is any point on the sphere, then what is $PA^2 + PB^2$ equal to?
(a) 15 (b) 14
(c) 13 (d) 6.5
[NDA – 2023 (1)]

Consider the following for the next two (02) items that follow:
Consider two lines whose direction ratios are (2, –1, 2) and (k, 3, 5). They are inclined at an angle $\pi/4$.

124. What is the value of k?
(a) 4 (b) 2
(c) 1 (d) –1
[NDA – 2023 (1)]

125. What are the direction ratios of a line which is perpendicular to both the lines?
(a) (1, 2, 10) (b) (–1, 2, 10)
(c) (11, 12, –10) (d) (11, 2, –10)
[NDA – 2023 (1)]

126. If L is the line with direction ratios <3, –2, 6 > and passing through (1, –1, 1), then what are the coordinates of the points on L whose distance from (1, –1, 1) is 2 units?
(a) $\left(-\frac{11}{7}, \frac{13}{7}, \frac{19}{7}\right)$ and $\left(\frac{1}{7}, \frac{3}{7}, \frac{5}{7}\right)$
(b) $\left(\frac{19}{7}, -\frac{11}{7}, \frac{18}{7}\right)$ and $\left(-\frac{1}{7}, \frac{3}{7}, -\frac{5}{7}\right)$
(c) $\left(\frac{13}{7}, \frac{11}{7}, \frac{19}{7}\right)$ and $\left(-\frac{1}{7}, -\frac{3}{7}, \frac{5}{7}\right)$
(d) $\left(\frac{13}{7}, -\frac{11}{7}, \frac{19}{7}\right)$ and $\left(\frac{1}{7}, -\frac{3}{7}, -\frac{5}{7}\right)$
[NDA-2023 (2)]

127. Which one of the planes is parallel to the line $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$?
(a) $2x + 2y + z - 1 = 0$ (b) $2x - y - 2z + 5 = 0$
(c) $2x + 2y - 2z + 1 = 0$ (d) $x - 2y + z - 1 = 0$
[NDA-2023 (2)]

128. What is the angle between the lines $2x = 3y = -z$ and $6x = -y = -4z$?
(a) 0° (b) 30°
(c) 60° (d) 90°
[NDA-2023 (2)]

129. What is the equation of the sphere concentric with the sphere $x^2 + y^2 + z^2 - 2x - 6y - 8z - 5 = 0$ and which passes through the origin?
(a) $x^2 + y^2 + z^2 - 2x - 8z = 0$
(b) $x^2 + y^2 + z^2 - 2x - 6y = 0$
(c) $x^2 + y^2 + z^2 - 6y - 8z = 0$
(d) $x^2 + y^2 + z^2 - 2x - 6y - 8z = 0$
[NDA-2023 (2)]

130. A point P lies on the line joining A(1,2,3) and B(2,10,1). If z-coordinate of P is 7, what is the sum of other two coordinates?
(a) –15 (b) –13
(c) –11 (d) –9
[NDA-2023 (2)]

131. If (1, –1, 2) and (2, 1, –1) are the end points of a diameter of a sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz - 1 = 0$, then what is u + v + w equal to ?
(a) –2 (b) –1
(c) 1 (d) 2
[NDA-2024 (1)]

132. If < l, m, n > are the direction cosines of a normal to the plane $2x - 3y + 6z + 4 = 0$, then what is the value of $49(7l^2 + m^2 - n^2)$?
(a) 0 (b) 1
(c) 3 (d) 71
[NDA-2024 (1)]

133. A line through (1, –1, 2) with direction ratios <3, 2, 2> meets the plane $x + 2y + 3z = 18$. What is the point of intersection of line and plane?
(a) (4, 4, 1) (b) (2, 4, 1)
(c) (4, 1, 4) (d) (3, 4, 7)
[NDA-2024 (1)]

134. If p is the perpendicular distance from origin to the plane passing through (1, 0, 0), (0, 1, 0) and (0, 0, 1), then what is $3p^2$ equal to?
(a) 4 (b) 3
(c) 2 (d) 1
[NDA-2024 (1)]

135. If the direction cosines < l, m, n > of a line are connected by relation $l + 2m + n = 0$, $2l - 2m + 3n = 0$, then what is the value of $l^2 + m^2 - n^2$?
(a) $\frac{1}{101}$ (b) $\frac{29}{101}$.
(a) $\frac{41}{101}$ (b) $\frac{92}{101}$
[NDA-2024 (1)]

Direction: Consider the following for the **two items** given below

Let $2x^2 + 2y^2 + 2z^2 + 3x + 3y + 3z - 6 = 0$ be a sphere.

136. What is the diameter of the sphere

(a) $\frac{5\sqrt{3}}{4}$ (b) $\frac{5\sqrt{3}}{2}$

(c) $\frac{3\sqrt{5}}{4}$ (d) $\frac{3\sqrt{5}}{2}$

[NDA-2024 (2)]

137. The centre of the sphere lies on the plane

(a) $2x + 2y + 2z - 3 = 0$ (b) $4x + 4y + 4z - 3 = 0$

(c) $4x + 8y + 8z - 15 = 0$ (d) $4x + 8y + 8z + 15 = 0$

[NDA-2024 (2)]

Direction: Consider the following for the **two items** given below

Let S be the line of intersection of two planes $x + y + z = 1$ and $2x + 3y - 4z = 8$.

138. Which of the following are the direction ratio of S ?

(a) $< -7, -6, 1 >$ (b) $< -7, 6, 1 >$

(c) $< -7, 5, 1 >$ (d) $< 6, 5, 1 >$

[NDA-2024 (2)]

139. If $< l, m, n >$ are direction cosines of S, then what is the value of $43(l^2 - m^2 - n^2)$?

(a) 6 (b) 5

(c) 4 (d) 1

[NDA-2024 (2)]

Direction: Consider the following for the **two items** given below

Let L: $x + y + z + 4 = 0 = 2x - y - z + 8$ be a line and P: $x + 2y + 3z + 1 = 0$ be a plane.

140. What are the direction ratio of the line ?

(a) $< 2, 1, -1 >$ (b) $< 0, -1, 2 >$

(c) $< 0, 1, -1 >$ (d) $< 2, 3, -3 >$

[NDA-2024 (2)]

141. What is the point of intersection of L and P ?

(a) (4, 3, –3) (b) (4, –3, 3)

(c) (–4, –3, –3) (d) (–4, –3, 3)

[NDA-2024 (2)]

142. If a line in 3 dimensions makes angles α, β, γ with the positive directions of the coordinate axes, then what is $\cos(\alpha+\beta)\cos(\alpha-\beta)$ equal to?

(a) $\cos^2\gamma$ (b) $-\cos^2\gamma$

(c) $\sin^2\gamma$ (d) $-\sin^2\gamma$

[NDA-2025 (1)]

143. A(1, 2, −1), B(2, 5, −2) and C(4, 4, −3) are three vertices of a rectangle. What is the area of the rectangle?

(a) 8 square units (b) 9 square units

(c) $\sqrt{66}$ square units (d) $\sqrt{68}$ square units

[NDA-2025 (1)]

144. ABC is a triangle right-angled at B. If A(k, 1, −1), B(2k, 0, 2) and C(2 + 2k, k, 1) are the vertices of the triangle, then what is the value of k?

(a) − 3 (b) − 1

(c) 1 (d) 3

[NDA-2025 (1)]

145. If a line $\frac{x+1}{p} = \frac{y-1}{q} = \frac{z-2}{r}$,

Where p = 2q = 3r, makes an angle θ with the positive direction of y-axis, then what is $\cos 2\theta$ equal to?

(a) −31/49 (b) −37/49

(c) 31/49 (d) 37/49

[NDA-2025 (1)]

146. What is the equation of the plane passing through the point (1, 1, 1) and perpendicular to the line whose direction ratios are $< 3, 2, 1 >$

(a) $x + 2y + 3z = 6$ (b) $3x + 2y + z = 6$

(c) $x + y + z = 3$ (d) $3x + 2y + z = 0$

[NDA-2025 (1)]

For the following two (02) items:

A plane P is parallel to the line having direction ratios ⟨1, 3, 2⟩ and contains the line of intersection of the planes $6x + 4y - 5z = 2$ and $x - 2y + 3z = 0$.

147. Which of the following are the direction ratios of the line of intersection of the given planes?

(a) ⟨2, 23, 16⟩ (b) ⟨2, −23, −16⟩

(c) ⟨2, 3, 2⟩ (d) ⟨−1, 3, −2⟩

[NDA-2025 (2)]

148. What is the equation of the plane P?

(a) $2x - 20y + 29z + 2 = 0$ (b) $2x - 20y + 29z - 2 = 0$

(c) $2x + 3y + 2z - 4 = 0$ (d) $x - 3y + 2z + 5 = 0$

[NDA-2025 (2)]

For the following two (02) items:

Suppose S is the sphere with the smallest radius that passes through the points

A(1, 0, 0), B(0, 1, 0) and C(0, 0, 1).

149. What is the radius of S?3

(a) $\sqrt{\frac{1}{3}}$ (b) $\sqrt{\frac{2}{3}}$

(c) $\frac{1}{3}$ (d) 1

[NDA-2025 (2)]

150. On which one of the following planes does the centre of S lie?

(a) $x + y + z - 1 = 0$ (b) $x + y + z + 1 = 0$

(c) $3x + 3y + 3z - 1 = 0$ (d) $3x + 3y + 3z + 1 = 0$

[NDA-2025 (2)]

For the following two (02) items:

Let A(1, −1, 0), B(−2, 1, 8) and C(−1, 2, 7) are three consecutive vertices of a parallelogram ABCD.

151. What is the fourth vertex D?

(a) (0, −2, 1) (b) (2, 0, −1)

(c) (1, 0, 1) (d) (1, 2, 0)

[NDA-2025 (2)]

152. If angle BCD is θ, then what is $\cos^2\theta$ equal to?

(a) 26/77 (b) 27/77

(c) 82/237 (d) 83/237

[NDA-2025 (2)]

ANSWER KEY

1.	a	**2.**	a	**3.**	a	**4.**	d	**5.**	b	**6.**	a	**7.**	b	**8.**	d	**9.**	b	**10.**	b
11.	b	**12.**	d	**13.**	b	**14.**	b	**15.**	c	**16.**	b	**17.**	c	**18.**	a	**19.**	a	**20.**	c
21.	a	**22**	a	**23.**	d	**24.**	d	**25.**	d	**26.**	b	**27.**	a	**28.**	c	**29.**	d	**30.**	b
31.	c	**32.**	a	**33.**	b	**34.**	c	**35.**	c	**36.**	c	**37.**	a	**38.**	d	**39.**	c	**40.**	c
41.	a	**42.**	a	**43.**	c	**44.**	b	**45.**	a	**46.**	a	**47.**	d	**48.**	d	**49.**	d	**50.**	a
51.	b	**52.**	c	**53.**	a	**54.**	d	**55.**	b	**56.**	c	**57.**	a	**58.**	b	**59.**	a	**60.**	b
61.	b	**62.**	a	**63.**	b	**64.**	c	**65.**	d	**66.**	c	**67.**	c	**68.**	a	**69.**	b	**70.**	c
71.	d	**72.**	c	**73.**	c	**74.**	b	**75.**	a	**76.**	a	**77.**	b	**78.**	c	**79.**	a	**80.**	d
81.	c	**82.**	a	**83.**	c	**84.**	c	**85.**	d	**86.**	c	**87.**	a	**88.**	b	**89**	d	**90.**	a
91.	a	**92.**	c	**93.**	b	**94.**	a	**95.**	b	**96.**	d	**97.**	a	**98.**	d	**99.**	b	**100.**	d
101.	c	**102.**	b	**103.**	b	**104.**	b	**105.**	c	**106.**	a	**107.**	c	**108.**	d	**109.**	c	**110.**	c
111.	c	**112.**	b	**113.**	b	**114.**	b	**115.**	d	**116.**	a	**117.**	b	**118.**	b	**119.**	b	**120.**	c
121.	c	**122.**	a	**123.**	b	**124.**	a	**125.**	d	**126.**	d	**127.**	d	**128.**	d	**129.**	d	**130.**	a
131.	a	**132.**	b	**133.**	c	**134.**	d	**135.**	b	**136.**	b	**137.**	d	**138.**	b	**139.**	a	**140.**	c
141.	d	**142.**	b	**143.**	c	**144.**	d	**145.**	a	**146.**	b	**147.**	b	**148.**	a	**149.**	b	**150.**	a
151.	b	**152.**	b																

Solutions

Sol.1. (a)
$a_1a_2 + b_1b_2 + c_1c_2 = 0$
lines are perpendicular.
Sol.2. (b)
radius = $\sqrt{\frac{1}{4}+\frac{1}{4}} = \frac{1}{\sqrt{2}}$
Sol.3. (a)
$\cos2\boldsymbol{\alpha} + \cos2\boldsymbol{\beta} + \cos2\boldsymbol{\gamma}$
$2\cos^2\boldsymbol{\alpha} - 1 + 2\cos^2 \; - 1 + 2\cos^2\boldsymbol{\gamma} - 1$
$2(\cos^2\boldsymbol{\alpha} + \cos^2\boldsymbol{\beta} + \cos^2\boldsymbol{\gamma}) - 3 = -1$
I is correct
$\cos^2\boldsymbol{\alpha} + \cos^2\boldsymbol{\beta} + \cos^2 \; = 1$
$1-\sin^2\boldsymbol{\alpha} + 1-\sin^2\boldsymbol{\beta} + 1-\sin^2\boldsymbol{\gamma} = 1$
$\sin^2\boldsymbol{\alpha} + \sin^2\boldsymbol{\beta} + \sin^2\boldsymbol{\gamma} = 2$
II is wrong.
Sol.4. (d)
$$\cos\theta = \left|\frac{1(-2)+1(1)+2(-1)}{\sqrt{1+1+4}\sqrt{4+1+1}}\right| = \frac{1}{2}$$
$\Rightarrow \boldsymbol{\theta} = 60^0$
Sol.5. (b)
only b option is satisfying the equation.
Sol.6. (a)
Sol.7. (b)
$$\cos\theta = \left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|$$
$$\cos\theta = \left|\frac{2(1)-1(1)+1(2)}{\sqrt{4+1+1}\sqrt{1+1+4}}\right| = \frac{1}{2}$$
$\Rightarrow \boldsymbol{\theta} = 60^0$
Sol.8. (d)
let the equation of the plane
$ax + by + cz + d = 0$
this is perpendicular to
$x-2y-8z = 0$
$2x+5y-z = 0$
so
$a-2b-8c = 0$
$2a+5b-c = 0$
$\frac{a}{42} = \frac{b}{-15} = \frac{c}{9}$
$a = 14, b = -5, c = 3$
$14x - 5y + 3z + d = 0$
$(1, -1, -1)$ lie on this plane
$14(1) - 5(-1) + 3(-1) + d = 0$
$d = -16$
then equation of plane
$14x - 5y + 3z = 16$
Sol.9. (b)
equation of sphere passing through origin
$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$
this passes through $(-1,0,0)$
so $1 - 2u = 0$
$u = 1/2$
and sphere passing through $(0,-2,0)$
so $4 - 4v = 0$
$v = 1$
and sphere passing through $(0,0,-3)$
so $9 - 6w = 0$
$w = 3/2$
equation of sphere is
$x^2 + y^2 + z^2 + x + 2y + 3z = 0$
Sol.10. (b)
D.R. of normal
$< 2, -1, 2 >$
or $< 1, -1/2, 1 >$
Sol.11. (b)
$$\cos\theta = \left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|$$
$$\cos\theta = \left|\frac{1(2)+1(-2)+1(2)}{\sqrt{1+1+1}\sqrt{4+4+4}}\right| = \frac{1}{3}$$
Sol.12. (c)
radius = $\sqrt{(2)^2+(-3)^2+(4)^2+7} = 6$
Sol.13. (c)
sum of square of direction cosine is always one.
Sol.14. (b)
$\mathbf{1 + \cos2\boldsymbol{\alpha} + \cos2\boldsymbol{\beta} + \cos2\boldsymbol{\gamma}}$
$\mathbf{1+2}\cos^2\boldsymbol{\alpha} - 1 + 2\cos^2\boldsymbol{\beta} - 1 + 2\cos^2\boldsymbol{\gamma} - 1$
$2(\cos^2\boldsymbol{\alpha} + \cos^2\boldsymbol{\beta} + \cos^2\boldsymbol{\gamma}) - 2 = 0$
Sol.15. (c)
$x = 3z + 4$, $y = 2z - 3$
$\frac{x-4}{3} = \frac{y+3}{2} = \frac{z}{1}$
Sol.16. (b)
Sol.17. (c)
equation of plane $3x + 4y - 5z + d = 0$
passes through $(1,2,3)$
$3(1) + 4(2) - 5(3) + d = 0$
$d = 4$
equation of plane $3x + 4y - 5z + 4 = 0$
Sol.18. (a)
distance of (x,y,z) from YZ plane is x coordinate of point.
Sol.19. (a)
in XZ plane y coordinate will be zero.
Sol.20. (c)
by distance formula
$\sqrt{(7-4)^2+(1-5)^2+(-3-\lambda)^2} = 13$
by solving it $\boldsymbol{\lambda} = 9$
Sol.21. (a)
$$\cos\theta = \left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|$$
$$\cos\theta = \left|\frac{1(1)-2(3/2)+1(2)}{\sqrt{1+4+1}\sqrt{1+9+4}}\right| = 0$$
lines are perpendicular.
Sol.22. (a)
$\cos^2\boldsymbol{\alpha} + \cos^2\boldsymbol{\beta} + \cos^2 \; = 1$
$\boldsymbol{\alpha} = \boldsymbol{\beta} = \; = 1$ (given)
$3\cos^2 \; = 1$
$\cos\boldsymbol{\alpha} = \frac{1}{\sqrt{3}}$
D.R. $< \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} >$
Sol.23. (d)
$$\cos\theta = \left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|$$
$$\cos\theta = \left|\frac{2(3)-1(-4)-2(5)}{\sqrt{4+1+4}\sqrt{9+16+25}}\right| = 0$$
planes are perpendicular
Sol.24. (d)
Sol.25. (d)
$$D = \left|\frac{d_1-d_2}{\sqrt{a^2+b^2+c^2}}\right|$$
$$D = \left|\frac{2-3}{\sqrt{9+36+9}}\right| = \frac{1}{\sqrt{54}}$$
Sol.26. (b)
coefficient of second order terms are same for sphere
$3 = k + 1$
$k = 2$
Sol.27. (a)
$\cos^2\boldsymbol{\alpha} + \cos^2\boldsymbol{\beta} + \cos^2 \; = 1$
$\frac{3}{4} + \cos^2\boldsymbol{\beta} + \cos^2 \; = 1$
$\cos^2\boldsymbol{\beta} + \cos^2 \; = \frac{1}{4}$
Sol.28. (c)
D.C. of z axis $< 0,0,1 >$
Sol.29. (d)
D.R. of one diagonal $< 1, 1, 1 >$
D.R. of another diagonal $< 1, -1, 1 >$
$$\cos\theta = \left|\frac{1-1+1}{\sqrt{1+1+1}\sqrt{1+1+1}}\right| = \frac{1}{3}$$
Sol.30. (b)
Sol.31. (c)
Sol.32. (a)
$$D = \left|\frac{ax_1+by_1+cz_1+d}{\sqrt{a^2+b^2+c^2}}\right|$$
$$D = \left|\frac{0+0+0-3}{\sqrt{4+1+4}}\right| = 1$$
Sol.33. (b)
length = $\sqrt{4+9+36} = 7$
Sol.34. (c)
centres $(0,2,0)$ and $(-1,0,-2)$
now distance between points by distance formula $= 3$
Sol.35. (c)
$r_1 = 1$ and $r_2 = 3$
$r_1 + r_2 >$ distance between centres
so sphere intersect each other
Sol.36. (c)
D.R. of line $<-4,4,2>$
Sol.37. (a)
Sol.38. (d)
Given point $A(1,-2,3)$
equation of line normal to the plane is
$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-1} = k$
any point on this line $P(2k+1, 3k-2, -k+3)$ ...(i)
this point also lie on plane
$2(2k+1)+3(3k-2)-1(-k+3) = 7$
$k = 1$
now by (i)
$P(3,1,2)$
Sol.39. (c)
P is mid point of A and its image (B point)
$B(5,4,1)$
Sol.40. (c)
$\alpha = 60$ and $\beta = 30$
we know that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
than $\gamma = 90$ [by above relation]
Sol.41. (a)
D.R. of line
$\left\langle \frac{1}{2}, \frac{\sqrt{3}}{2}, 0 \right\rangle$
Sol.42. (a)
equation of plane passing through three points is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x-x_2 & y-y_2 & z-z_2 \\ x-x_2 & y-y_2 & z-z_3 \end{vmatrix} = 0$$

so plane passing through (2,2,1) , (3,4,2) , (7,0,6)

$$\begin{vmatrix} x-2 & y-2 & z-1 \\ 1 & 2 & 1 \\ 5 & -2 & 5 \end{vmatrix} = 0$$

by solving it

$x-z = 1$

this plane passes through (1,0,0)

Sol.43. (c)

D.R. of normal of above plane < 1,0,−1 >

Sol.44. (b)

Equation of sphere passes through origin

$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$

which passes through the points (2,1,−1), (1,5,−4), and (−2,4,−6).

so $4u + 2v - 2w = -6$

$2u + 10v - 8w = -42$

$-4u + 8v - 12w = -56$

from solving these equations

$u = 1, v = -2, w = 3$

radius of sphere = $\sqrt{1+4+9} = \sqrt{14}$

Sol.45. (a)

centre by above solution (−u, −v, −w) (−1,2, −3)

Sol.46. (a)

$x^2 + y^2 + z^2 + 2x - 4y + 6z = 0$

this sphere passes through (0,4,0)

distance of (1,1,1) from centre of sphere (−1,2, −3) is 3

Sol.47. (d)

let plane cut in k:1

then coordinate of intersection point

$$\left(\frac{4k+2}{k+1}, \frac{-2k+1}{k+1}, \frac{5k+3}{k+1}\right)$$

this point will lie on plane so

$$2\left(\frac{4k+2}{k+1}\right) + \left(\frac{-2k+1}{k+1}\right) - \left(\frac{5k+3}{k+1}\right) = 3$$

by solving it k = − 1/2

means plane intersect externally in ratio 1:2

put this value of k in intersection point

then point (0,4,1)

Sol.48. (d)

by above solution plane intersect externally in ratio 1:2

Sol.49. (d)

length of line segment = $\sqrt{(12)^2 + (4)^2 + (3)^2} = 13$ units

Sol.50. (a)

Direction cosines = $\left(\frac{12}{13}, \frac{4}{13}, \frac{3}{13}\right)$

Sol.51. (b)

let Q is foot of perpendicular that lies on line

$$\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4} = k$$

then Q(2k , 3k + 2, 4k +3)

D.R. of PQ < 2k−3, 3k + 3, 4k−8 >

PQ line and given line is perpendicular to each other

$2(2k-3) + 3(3k+3) + 4(4k-8) = 0$

$29k = 29$

$k = 1$

Then D.R. of PQ < −1, 6 ,−4 >

Sol.52. (c)

coordinate of Q(2,5,7)

Distance between P and Q by distance formula

$$\sqrt{(2-3)^2 + (5+1)^2 + (7-11)^2} = \sqrt{53}$$

Sol.53. (a)

Let P(x,0,0) on X axis

Q(0,y,0) on Y axis

R(0,0,z) on Z axis

centroid $\frac{x+0+0}{3} = 1 \Rightarrow x=3$

similarly $\frac{0+y+0}{3} = 2 \Rightarrow y = 6$

and $\frac{0+0+z}{3} = 3 \Rightarrow z = 9$

Sol.54. (d)

Equation of plane

$$\frac{x}{3} + \frac{y}{6} + \frac{z}{9} = 1$$

$6x + 3y + 2 = 18$

Sol.55. (b)

length of diagonal

$= \sqrt{(1)^2 + (2)^2 + (3)^2} = \sqrt{14}$ units

Sol.56. (c)

Equation of plane parallel to XY plane is z = c

Sol.57. (a)

$$\frac{x}{1/2} = \frac{y}{1/3} = \frac{z}{-1}, \frac{x}{1/6} = \frac{y}{-1} = \frac{z}{-1/4}$$

for these lines $a_1a_2 + b_1b_2 + c_1c_2 = 0$

so lines are perpendicular.

Sol.58. (b)

given sphere

$$x^2 + y^2 + z^2 - \frac{8}{3}x + \frac{4}{3}y + \frac{8}{3}z - 5 = 0$$

radius= $\sqrt{\frac{16}{9} + \frac{4}{9} + \frac{16}{9} + 5} = 3$

Sol.59. (a)

Let D.R. of line < a,b,c >

this is perpendicular to both

so $a - 2b - 2c = 0$

and $2b + c = 0$

$$\frac{a}{2} = \frac{b}{-1} = \frac{c}{2}$$

D.R. of line < 2,−1, 2 >

Sol.60. (b)

at XY plane z = 0

Sol.61. (b)

$5x + 2y + z = 13$

$$\frac{x}{13/5} + \frac{y}{13/2} + \frac{z}{13} = 1$$

Sol.62. (a)

Let D.R. of that line are < a , b , c >

and this line is intersection lines of given planes

$2x - y + 3z = 2$ and $x + y - z = 1$

so

$2a - b + 3c = 0$

$a + b - c = 0$

$$\frac{a}{2} = \frac{b}{-5} = \frac{c}{-3}$$

D.R. < 2,−5,−3>

Sol.63. (b)

equation of plane passing through the intersection point of given planes is

$2x - y + 3z - 2 + \lambda(x + y - z - 7) = 0$

this plane passes through (1,0,1)

$2(1) - (0) + 3(1) - 2 + \lambda(1 + 0 - 1 - 7) = 0$

$\lambda = 3$

so equation of plane

$5x + 2y - 5 = 0$

Sol.64. (c)

plane P touches the sphere $x^2 + y^2 + z^2 = r^2$

r = Distance between centre of sphere (0,0,0) to place P.

$$r = \left|\frac{5(0) + 2(0) - 5}{\sqrt{5^2 + 2^2 + 0}}\right| = \frac{5}{\sqrt{29}}$$

Sol.65. (d)

Let $Q(x_1, y_1, z_1)$ be the image of the point P.

D.R. of PQ are 3,−2,2 ....(i)

The equation of line PQ is

$$\frac{x+2}{3} = \frac{y-1}{-2} = \frac{z+5}{2} = r$$

Coordinate o0f any point on the line PQ is 3r−2,−2r+1,2r−5.

Let Q (3r−2,−2r+1,2r−5) be such a point

Let L be the mid point of PQ,

$$L = \left(\frac{3r}{2} - 2, 1 - r, r - 5\right)$$

Since L lies on the plane $3x - 2y + 2z + 1 = 0$

So, $3\left(\frac{3r}{2} - 2\right) - 2(1 - r) + 2(r - 5) + 1 = 0$

r = 2

then Q(4,−3,−1) .....(ii)

Also the mid point of PQ is L (1,−1,−3).....(iii)

so by distance formula PQ = $2\sqrt{17} > 8$

option d is correct.

Sol.66. (c)

from equation (i) of above solution.

Sol.67. (c)

Let a,b,c be the direction ratios of the lines.

Then the equation is

$$\frac{x-5}{a} = \frac{y+6}{b} = \frac{z-7}{c} \quad ....(i)$$

Since (i) is parallel to the planes $x + y + z = 1$ and $2x - y - 2z = 3$ then

$a(1) + b(1) + c(1) = 0$

and $a(2) + b(-1) + c(-2) = 0$

By cross multiplication

$$\frac{a}{-1} = \frac{b}{4} = \frac{c}{-3}$$

D.R. < −1,4,−3> = < 1,−4, 3 >

Sol.68. (a)

Substituting a,b,c in (i) we get

$$\frac{x-5}{-1} = \frac{y+6}{4} = \frac{z-7}{-3}$$

Sol.69. (b)

Sol.70. (c)

A(0,0,0) , B(a,0,0) , C(0,b,0) ,D(0,0,c)

Let the equidistant point be P(x,y,z)

i.e, AP = BP, AP = CP, AP = DP

$\Rightarrow \sqrt{(x)^2 + (y)^2 + (z)^2} = \sqrt{(x-a)^2 + (y)^2 + (z)^2}$

squaring both sides then after solving it

x =a/2

Similarly y =b/2 , z =c/2

Sol.71. (d)

equation of line by joining these points is

$$\frac{x-1}{3-1} = \frac{y-2}{-1-2} = \frac{z+1}{2+1}$$

on YZ plane x = 0

put x = 0 in above line then x = 7/2 , y = −5/2

Sol.72. (c)

Given lines

x = ay + b, and z = cy + d

$$\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c}$$

Also x = ey + f, and z = gy + h

$$\frac{x-f}{e} = \frac{y}{1} = \frac{z-h}{g}$$

these lines are perpendicular
$a_1a_2 + b_1b_2 + c_1c_2 = 0$
ae + 1 + cg = 0
ae + cg + 1 = 0

Sol.73. (c)
P(3,2,4) , Q(4,5,2) , R(5,8,0), S(2,-1,6)
by distance formula
PQ = $\sqrt{14}$, QR = $\sqrt{14}$,RS = 3 $\sqrt{14}$,PS = $\sqrt{14}$
Since, PQ + QR + RS + PS = RS
points are collinear.

Sol.74. (b)

$$D=\left|\frac{ax_1+by_1+cz_1+d}{\sqrt{a^2+b^2+c^2}}\right|$$

$$D=\left|\frac{0+0+0-9}{\sqrt{1+4+4}}\right|=3$$

Sol.75. (a)
equation of line by joining these points is

$$\frac{x+3}{5+3}=\frac{y-4}{-6-4}=\frac{z+8}{12}$$

on XY plane z = 0
put z = 0 in above line then x = 7/3 , y = −8/3

Sol.76. (a)
equation of plane passing through the intersection point of given planes is
x + y + z −1 + λ(2x + 3y + 4z−7) = 0
x(1+2λ) + y(1+3λ) + z(1+4λ) −1−7λ = 0
this plane is perpendicular to
x − 5y + 3z − 5 = 0
then $a_1a_2 + b_1b_2 + c_1c_2 = 0$
1(1+2λ) − 5(1+3λ) +3(1+4λ) = 0
λ = −1
equation of plane
x + y + z −1 + (−1)(2x + 3y + 4z−7) = 0
x + 2y + 3z −6 = 0

Sol.77. (b)
D.R. of lines are < 2,−1,2 > and < x,3,5 >angle between lines is 45^0

$$\cos\theta=\left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|$$

$$\cos\frac{\pi}{4}=\left|\frac{2(x)-1(3)+2(5)}{\sqrt{4+1+4}\sqrt{x^2+9+25}}\right|$$

$$\frac{1}{\sqrt{2}}=\left|\frac{2x+7}{3\sqrt{x^2+34}}\right|$$

squaring both sides

$$\frac{1}{2}=\frac{(2x+7)^2}{9(x^2+34)}$$

x^2 −56x + 208 = 0
x = 4, 52
smaller value = 4

Sol.78. (c)
Let plane cuts the x axis at (p,0,0), Y axis at (0,q,0) , Z axis at (0,0,r)
equation of plane is $\frac{x}{p}+\frac{y}{q}+\frac{z}{r}=1$
this plane passes through (a,b,c)
so $\frac{a}{p}+\frac{b}{q}+\frac{c}{r}=1$
Equation of sphere is
$x^2 + y^2 + z^2$ −px −qy −rz = 0
centre of sphere is (l,m,n) = $\left(\frac{p}{2},\frac{q}{2},\frac{r}{2}\right)$
p = 2l, q = 2m, r = 2n
locus of centre is

$$\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=2$$

Sol.79. (a)
equation of sphere
$(x + 2)^2 + (y − 3)^2 + (z − 4)^2 = 36$
$x^2 + y^2 + z^2$ + 4x −6y −8z = 7

Sol.80. (d)
P(1,−1,1) , Q(3,−2,2) , R(0,2,6)
D.R. of PQ = <2,−1,1>
D.R. of PR = <−1,3,5>

$$\cos P=\left|\frac{2(-1)-1(3)+1(5)}{\sqrt{2^2+(-1)^2+1^2}\sqrt{(-1)^2+3^2+5^2}}\right|=0$$

∠P = 90^0
∠Q = θ
∠R = 90^0 − θ

Sol.81. (c)
let the sphere passing through points A(a,0,0) , B(0,0,b) , C(0,0,c)
Equation of sphere is
$x^2 + y^2 + z^2$ −ax −by −cz = 0
radius, r = $\frac{1}{2}\sqrt{a^2+b^2+c^2}$
⇒ $a^2 + b^2 + c^2 = 4r^2$
Let (α,β,γ) be centroid of sphere.
(α,β,γ)
= $\left(\frac{a+0+0}{3},\frac{0+b+0}{3},\frac{0+0+c}{3}\right)$
= $\left(\frac{a}{3},\frac{b}{3},\frac{c}{3}\right)$
$\alpha^2+\beta^2+\gamma^2=\left(\frac{a^2}{9}+\frac{b^2}{9}+\frac{c^2}{9}\right)$
$\alpha^2+\beta^2+\gamma^2=\left(\frac{4r^2}{9}\right)$
$9(\alpha^2+\beta^2+\gamma^2) = 4r^2$
Locus is $9(x^2 + y^2 + z^2) = 4r^2$

Sol.82. (a)
equation of plane passing through three points is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x-x_2 & y-y_2 & z-z_2 \\ x-x_2 & y-y_2 & z-z_3 \end{vmatrix}=0$$

so plane passing through (-2,6,-6) , (-3,10,-9) , (-5,0,-6)

$$\begin{vmatrix} x+2 & y-6 & z+6 \\ -1 & 4 & -3 \\ -3 & -6 & 0 \end{vmatrix}=0$$

by solving it
2x− y−2z = 2

Sol.83. (c)
equation of line BC

$$\frac{x-0}{2}=\frac{y+11}{8}=\frac{z-4}{-3}$$

let any point P is foot of perpendicular

$$\frac{x-0}{2}=\frac{y+11}{8}=\frac{z-4}{-3}=k$$

x = 2k , y = 8k−11, z = −3k + 4
P(2k , 8k−11, −3k + 4)
D.R of AP < 2k−1, 8k−19 , −3k >
AP and BC is perpendicular
so 2(2k−1)+8(8k−19) -3(−3k) = 0
k = 2
hence P(4,5,−2)

Sol.84. (c)

$$\cos\theta=\left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|$$

given planes 2x − y + z = 1, x + y + 2z=3

$$\cos\theta=\left|\frac{(2)(1)+(-1)(1)+(1)(2)}{\sqrt{2^2+(-1)^2+1^2}\sqrt{1^2+1^2+2^2}}\right|$$

$$\cos\theta=\frac{1}{2}$$

angle = 60^0
I statement is correct

$$D=\left|\frac{d_1-d_2}{\sqrt{a^2+b^2+c^2}}\right|$$

$$D=\left|\frac{12-2}{\sqrt{6^2+3^2+6^2}}\right|=\frac{10}{9}$$

both statements are correct.

Sol.85. (d)
drs of line is 2,3,4
going through options
2(1) + 3(2) + 4(-2) = 0
2(4) + 3(4) + 4(-5) = 0

Sol.86. (c)
going through option (c)

Sol.87. (a)

$$D=\left|\frac{ax_1+by_1+cz_1+d}{\sqrt{a^2+b^2+c^2}}\right|$$

$$D=\left|\frac{3(2)-6(3)+2(4)+11}{\sqrt{9+36+4}}\right|=1$$

Sol.88. (b)
Sol.89. (d)
Sol.90. (a)

$$D=\left|\frac{d_1-d_2}{\sqrt{a^2+b^2+c^2}}\right|$$

$$D=\left|\frac{21-18}{\sqrt{8^2+(-4)^2+8^2}}\right|=\frac{3}{12}=\frac{1}{4}$$

Sol.91. (a)
2x + y + 2z = 9, 4x − 5y − 4z = 1
equation of family of plane
(2x + y + 2z − 9) +λ(4x − 5y − 4z − 1) = 0
point (3,2,1) lie on this plane so put this point in above plane
(2(3) + 2 + 2 − 9) +λ(4(3) − 5(2) − 4 − 1) = 0
⟹ λ = 1/3
(2x + y + 2z − 9) +1/3 (4x − 5y − 4z − 1) = 0
10x −2y + 2z = 28

Sol.92. (c)
equation of sphere
$x^2 + y^2 + z^2$ + 2ux + 2vy + 2wz + c = 0
radius = $\sqrt{u^2+v^2+w^2-c}$
here u = 3 , v = −4 , w = 5 , c = 1
radius = $\sqrt{9+16+25-1}=7$

Sol.93. (b)

$$l=k,m=\sqrt{3}k,n=\cos 60^0$$

$$l^2+m^2+n^2=1$$

$k^2 + 3k^2 + \frac{1}{4}=1$
$4k^2 = \frac{3}{4}$, k = $\pm\frac{\sqrt{3}}{4}$
Possible D.C. $<\frac{\sqrt{3}}{4},\frac{3}{4},\frac{1}{2}>$
and $<-\frac{\sqrt{3}}{4},-\frac{3}{4},\frac{1}{2}>$

$$\cos\theta=l_1l_2+m_1m_2+n_1n_2$$

$\cos\theta = -\frac{\sqrt{3}}{4}\left(\frac{\sqrt{3}}{4}\right) - \frac{3}{4}\left(\frac{3}{4}\right) + \frac{1}{2}\left(\frac{1}{2}\right)$

$\cos\theta = -\frac{1}{2}$

$\theta = 120^0$ then acute angle $= 60^0$

Sol.94. (a)
Let p = 1
then $A(2,-2,\sqrt{2})$
now put p = 2
then $B(3,-1, 2\sqrt{2})$
D.R. of AB = $1,1, \sqrt{2}$
then Direction cosine $< \frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} >$

Sol.95. (b)
only point (2,5,5) satisfies the given equation.

Sol.96. (d)
If line lie on plane then point (4,2,k) will also lie on plane.
$2(4) - 4(2) + k = 7 \Rightarrow k = 7$

Sol.97. (a)
$A(x,y,-3)$, $B(2,0,-1)$, $C(4,2,3)$
D.R. of AB = $< 2-x , -y , 2 >$
D.R. of BC = $< 2 , 2 , 4 >$
if these points are collinear then D.R. will be in same ratio
$\frac{2-x}{2} = \frac{-y}{2} = \frac{2}{4}$
$x = 1, y = -1$

Sol.98. (d)
radius of sphere = perpendicular distance from centre of sphere to plane.
$= \frac{6(1)-3(-2)+2(3)-4}{\sqrt{36+9+4}} = \frac{14}{7} = 2$
then diameter = 4 units

Sol.99. (b)
$\frac{x-0}{1} = \frac{y-0}{0} = \frac{z-0}{0}$?
given equation is represents X axis
so distance of (2,3,4) from X axis is $\sqrt{3^2+4^2} = 5$ units

Sol.100.(d)
sum of squares of direction cosines of any line is always 1

Sol.101.(c)
coordinates planes divide the space in 8 compartments that is called octants.

Sol.102.(b)
equation of xy plane is z = 0
and equation of plane parallel to it at 5 distance is z = 5

Sol.103.(b)
direction ratio {6,3,6} and {3,3,0}
$\text{Cos}\,\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$
$\text{Cos}\,\theta = \frac{(6)(3)+(3)(3)+(6)(0)}{\sqrt{36+9+36}\sqrt{9+9+0}}$
$\text{Cos}\,\theta = \frac{18+9}{9\times\sqrt{18}} = \frac{1}{\sqrt{2}}$
$\theta = 45°$

Sol.104. (b)
$x-1 = 2(y+3) = 1 - z,$
$\frac{(x-1)}{2} = \frac{(y+3)}{1} = \frac{(z-1)}{-2}$
D.R = $< 2, 1, -2 >$
D.C. = $< l, m, n >$ = $< \frac{2}{3}, \frac{1}{3}, \frac{-2}{3} >$

$l^4 + m^4 + n^4 = \left(\frac{2}{3}\right)^4 + \left(\frac{1}{3}\right)^4 + \left(\frac{-2}{3}\right)^4$

$\frac{16}{81} + \frac{1}{81} + \frac{16}{81} = \frac{11}{27}$

Sol.105. (c)
projection of the line segment joining A(1, 7, –5) and B(–3, 4, –2) on y-axis
D.R. of line = $< -4, -3, 3 >$
Projection on Y axis = $r\cos\beta = |-3| = 3$

Sol.106.(a)
points A(k, 1, 3) and B(1, –2, k + 1) also passes through the point (15, 2 –4)?
equation of line AB
$= \frac{x-k}{1-k} = \frac{y-1}{-2-1} = \frac{z-3}{k+1-3}$
passes through the point (15, 2 –4)?
$\frac{15-k}{1-k} = \frac{2-1}{-2-1} = \frac{-4-3}{k+1-3}$
by solving any two equations we find different values of k, which not satisfy whole equation.

Sol.107.(c)
D.R. of line perpendicular to plane is <1,1,1>
then equation of line passing through (0,0,0) whose D.R. <1, 1, 1> is
x = y = z
point on this line x = y = z =k or (k, k, k)
this point also lies on plane so
k + k + k = 3
k = 1
then foot of perpendicular is (1,1,1)

Sol.108.(d)
Z= k is the equation of plane parallel to xy – plane

Sol.109.(c)
a : b : c
are direction ratio it may be infinite times
$\ell^2 + m^2 + n^2 = \ell + m + n$
This is possible
For may lines.
Later x-axis, y-axis, z-axis
Both statements are correct.

Sol.110.(c)
On xy-plane z coordinate is zero

[redacted]

$O = \frac{m(6)+n(4)}{m+n}$
6m + 4x = 0
6m = – 44
$\frac{m}{n} = -\frac{2}{3}$
Divide externally in Ratio 2 : 3

Sol.111.(c)
One sphere will lie in each octant.

Sol.112.(b)

[redacted]

D.R. of AB <12, 0, 0 >
D.R. of AG <12, 6, 4>
Angle between AB and AG
$\cos\alpha = \frac{12(12)+6(0)+4(0)}{12\times\sqrt{12^2+6^2+4^2}}$
$= \frac{12\times12}{12\times14} = \frac{6}{7}$
D.R. & AC < 12, 6, 0, >
D.R. of A.G. <12, 6, 47
Angle between AC and AG
$\cos\beta = \frac{12\times12+6\times6+6\times4}{\sqrt{12^2+6^2+0^2}\sqrt{12^2+6^2+4^2}}$
$= \frac{180}{\sqrt{180}\sqrt{196}} = \sqrt{\frac{180}{196}} = \sqrt{\frac{45}{49}}$
$\cos2\alpha + \cos2\beta = 2\cos^2\alpha - 1 + 2\cos^2\beta - 1$
$= 2\left(\frac{36}{49}\right) - 1 + 2\left(\frac{45}{49}\right) - 1$
$= 2\left(\frac{81}{49}\right) - 2 = \frac{162-98}{49} = \frac{64}{49}$

Sol.113.(b)
sphere $x^2 + y^2 + z^2 - 2x - 3y - 4z = 0$
cuts the x axis at A (2, 0, 0)
and y axis at B (0, 0, 4)
equation of plane that makes OA, OB, OC intercepts on axes
$\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$`
6x + 4y + 3z – 12 = 0
By comparing with 6x + ky + 3z – 12,
K=4

Sol.114.(b)
Centre of given sphere is (1, 3/2, 2)
Perpendicular distance from (1, 3/2, 2) to the plane
6x + 4y + 3z –12 = 0
$P = \left|\frac{6(1)+4\left(\frac{3}{2}\right)+3(2)-12}{\sqrt{36+16+9}}\right|$
$P = \frac{6}{7} = 0.86$
0.5 < P < 1

Sol.115.(d)
Origin (0, 0, 0)
Centre of sphere $(1, \frac{3}{2}, 2)$
Equation of line will be
$\Rightarrow$ 6x –=4y = 3z

Sol.116.(a)
$\frac{2x}{k} + \frac{2y}{3} + \frac{2}{3} = 2$
Pass through the point (2, 3) –6)
$\frac{2(2)}{k} + \frac{2(3)}{3} + \frac{(-6)}{3} = 2$
$\frac{4}{k} + 2 - 2 = 2$
K=2
So eqn of plane is
$x + \frac{2y}{3} + \frac{z}{3} = 2$
3x + 2y + z – 6 = 0
Direction ratio < 3, 2, 1>

Sol.117.(b)
3x + 2y + 2 = 6
$\frac{x}{2} + \frac{y}{3} + \frac{z}{6} = 1$
p+q+r = 2 + 3 + 6 = 11

Sol.118.(b)
D.R. of line AB < 1 , 2, 1 > and D.R. of line CD < 1, 2, 2 > . so lines are not parallel. Rectangle answer is not possible.
mid point of AC and BD is (3, 5, 5)

Sol.119.(b)
$x^2 + y^2 + z^2 - 4x - 6y - 8z - 16 = 0$
centre of sphere (2, 3, 4)
this centre satisfies the equation $x + y + z - 9 = 0$
so statement 2 is correct.
radius of sphere $= \sqrt{4+9+16+16} = \sqrt{45}$ that is not equal to distance of centre from z axis (i.e. $\sqrt{4+9} = \sqrt{13}$) so statement 1 is wrong

Sol.120.(c)
equation of plane by intercept form
$$\frac{x}{2} + \frac{y}{2} + \frac{z}{1} = 1$$
or $x + y + 2z - 2 = 0$
D.R. of perpendicular line of plane < 1, 1, 2>
D.C. will be $< \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} >$

Sol.121.(c)
D.R. of y axis is < 0, k, 0 >
D.R. of z axis is < 0, 0, k > that is perpendicular to < 5, 6, 0 >
because $a_1a_2 + b_1b_2 + c_1c_2 = 0$

Sol.122.(a)
If A(1,–1, 2) and B(2, 1, –1) be the end points of the diameter of the sphere then equation of sphere is
$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$
$(x - 1)(x - 2) + (y + 1)(y - 1) + (z - 2)(z + 1) = 0$
$x^2 + y^2 + z^2 - 3x - z - 1 = 0$
compare it with $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz - 1 = 0$
$u = -3/2$, $v = 0$, $w = -½$
$u + v + w = -2$

Sol.123.(b)
If P (x, y, z) is any point on the sphere, then triangle PAB is a right angled triangle, and diameter of sphere will be hypogenous of triangle
$PA^2 + PB^2 = AB^2$
Distance between A(1,–1, 2) and B(2, 1, –1)
AB = $\sqrt{(2-1)^2 + (1+1)^2 + (-1-2)^2} = \sqrt{14}$
$AB^2 = 14$units

Sol.124.(a)
two lines whose direction ratios are (2, –1, 2) and (k, 3, 5), They are inclined at an angle $\pi/4$
$$\cos\theta = \left|\frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}\right|$$
$$\cos\frac{\pi}{4} = \left|\frac{2k - 3 + 10}{\sqrt{9}\sqrt{k^2 + 9 + 25}}\right|$$
$$\frac{1}{2} = \left|\frac{(2k+7)^2}{9(k^2+34)}\right|$$
$9k^2 + 306 = 8k^2 + 56k + 98$
$k^2 - 56k + 208 = 0$
k = 52, 4

Sol.125.(d)
by above solution D.R. of both lines will be (2, –1, 2) and (4, 3, 5)
let < a, b, c > are D.R. of a line perpendicular to both above lines so
$2a - b + 2c = 0$ …….(i)
$4a + 3b + 5c = 0$ …….(ii)
by eq (i) and (ii)
$$\frac{a}{-5-6} = \frac{b}{8-10} = \frac{c}{6+4}$$
$$\frac{a}{-11} = \frac{b}{-2} = \frac{c}{10}$$
so < a, b, c > will be < – 11, – 2, 10 >
or < 11, 2 , – 10 >

Sol.126.(d)
L is the line with direction ratios <3, –2, 6 > and passing through (1, –1, 1)
equation of line
$$\frac{x-1}{3} = \frac{y+1}{-2} = \frac{z-1}{6}$$
let any point on this line is (3k + 1, – 2k – 1, 6k + 1), distance of this point from (1, –1, 1) is 2 units
$$\sqrt{9k^2 + 4k^2 + 36k^2} = 2$$
so k = 2/7 or – 2/7
put these values of k in assumed point then we have $\left(\frac{13}{7}, -\frac{11}{7}, \frac{19}{7}\right)$ and $\left(\frac{1}{7}, -\frac{3}{7}, -\frac{5}{7}\right)$

Sol.127.(d)
If a line and a plane is parallel than
$a_1a_2 + b_1b_2 + c_1c_2 = 0$
by options (iv)
$x - 2y + z - 1 = 0$ is parallel to $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$, because
$3(1) + 4(-2) + 5(1) = 0$

Sol.128.(d)
angle between the lines
$\frac{x}{1/2} = \frac{y}{1/3} = \frac{z}{-1}$ and $\frac{x}{1/6} = \frac{y}{-1} = \frac{z}{-1/4}$
$$\cos\theta = \left|\frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}\right|$$
for above equations
$a_1a_2 + b_1b_2 + c_1c_2 = 0$
so lines are perpendicular.

Sol.129.(d)
equation of the sphere concentric with the sphere
$x^2 + y^2 + z^2 - 2x - 6y - 8z - 5 = 0$ is
$x^2 + y^2 + z^2 - 2x - 6y - 8z + k = 0$
if above sphere passing through origin than k = 0
so equation of sphere is
$x^2 + y^2 + z^2 - 2x - 6y - 8z = 0$

Sol.130.(a)
equation of line passing through A(1,2,3) and B(2,10,1) is
$$\frac{x-1}{1} = \frac{y-2}{8} = \frac{z-3}{-2}$$
a point P(x, y, 7) lie on above line
$$\frac{x-1}{1} = \frac{y-2}{8} = \frac{7-3}{-2}$$
x = – 1 , y = – 14
x + y = – 15

Sol.131.(a)
If A(1,–1, 2) and B(2, 1, –1) be the end points of the diameter of the sphere then equation of sphere is
$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$
$(x - 1)(x - 2) + (y + 1)(y - 1) + (z - 2)(z + 1) = 0$
$x^2 + y^2 + z^2 - 3x - z - 1 = 0$
compare it with $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz - 1 = 0$
$u = -3/2$, $v = 0$, $w = -½$
$u + v + w = -2$

Sol.132.(b)
D.R. of normal of plane is < 2, –3, 6 >
than direction cosine < l,m,n > will be $< \frac{2}{7}, -\frac{3}{7}, \frac{6}{7} >$
$49(7l^2 + m^2 - n^2)$
$$= 49\left(7 \times \frac{4}{49} + \frac{9}{49} - \frac{36}{49}\right) = 1$$

Sol.133.(c)
A line through (1, –1, 2) with direction ratios < 3, 2, 2 > will be
$$\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-2}{2}$$
let P is an intersection point of line and plane than
$$\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-2}{2} = k$$
coordinates of P are (3k + 1, 2k – 1, 2k + 2)
this point also lie on plane , so this will satisfy the equation of plane
$(3k + 1) + 2(2k - 1) + 3(2k + 2) = 18$
13k = 13
k = 1
coordinates of P are (4, 1, 4)

Sol.134.(d)
plane is passing through (1, 0, 0), (0, 1, 0) and (0, 0, 1), it means x, y, z intercepts are 1,1,1
so equation of plane by intercept form is
$$\frac{x}{1} + \frac{y}{1} + \frac{z}{1} = 1$$
x + y + z = 1 or x + y + z – 1 = 0
perpendicular distance of above plane from origin is
$$p = \left|\frac{0+0+0-1}{\sqrt{1+1+1}}\right| = \frac{1}{\sqrt{3}}$$
$3p^2 = 1$

Sol.135.(b)
$l + 2m + n = 0$ …………(i)
$2l - 2m + 3n = 0$ ………..(ii)
by adding both equations
$3l + 4n = 0$
$l / n = -4/3$
let $l = -4k$ and n = 3k , put these values in (i)
– 4k + 2m + 3k = 0
m = k/2
we know that $l^2 + m^2 + n^2 = 1$
$16k^2 + k^2/4 + 9k^2 = 1$
$101k^2 = 4$
$k^2 = 4/101$
$l^2 + m^2 - n^2 = 16k^2 + k^2/4 - 9k^2 = 29k^2/4$
put the value of k^2
$29k^2/4 = 29/101$

Sol.136.(b)
$2x^2 + 2y^2 + 2z^2 + 3x + 3y + 3z - 6 = 0$
$x^2 + y^2 + z^2 + 3/2x + 3/2y + 3/2z - 3 = 0$
coordinates of centre $\left(-\frac{3}{4}, -\frac{3}{4}, -\frac{3}{4}\right)$
radius = $\sqrt{\frac{9}{16} + \frac{9}{16} + \frac{9}{16} + 3} = \frac{\sqrt{75}}{4} = \frac{5\sqrt{3}}{4}$
diameter = $\frac{5\sqrt{3}}{2}$

Sol.137.(d)
coordinates of centre $\left(-\frac{3}{4},-\frac{3}{4},-\frac{3}{4}\right)$ satisfying the equation 4x + 8y + 8z + 15 = 0
Sol.138.(b)
let D.R. of line S are < a, b, c >
D.R. of line A perpendicular to x + y + z = 1 is < 1, 1, 1 >
D.R. of line B perpendicular to 2x + 3y – 4z = 1 is < 2, 3, – 4 >
line S and A are perpendicular so a + b + c = 0
line S and B are perpendicular so 2a+ 3b – 4c = 0
by solving both equations
$$\frac{a}{-4-3}=\frac{b}{2+4}=\frac{c}{3-2}$$
D.R. of line S are < –7, 6, 1 >
Sol.139.(a)
by above solution
D.R. of line S are < –7, 6, 1 >
D.C. of line S are $<\frac{-7}{\sqrt{86}},\frac{6}{\sqrt{86}},\frac{1}{\sqrt{86}}>$
$43(l^2 - m^2 - n^2) = 43\left(\frac{49}{86}-\frac{36}{86}-\frac{1}{86}\right)=6$
Sol.140.(c)
let D.R. of line L are < a, b, c >
D.R. of line A perpendicular to x + y + z + 4 = 0 is < 1, 1, 1 >
D.R. of line B perpendicular to 2x – y – z + 8 = 0 is < 2, –1, – 1 >
line L and A are perpendicular so a + b + c = 0
line L and B are perpendicular so 2a –b – c = 0
by solving both equations
$$\frac{a}{-1+1}=\frac{b}{2+1}=\frac{c}{-1-2}$$
D.R. of line L are < 0, 3, –3 > or < 0, 1, –1 >
Sol.141.(d)
let any common point on both planes is Q whose z coordinate is zero.
so x + y + 4 = 0 and 2x – y + 8 = 0
by solving above equations we get x = – 4 , y = 0
so coordinate of Q(– 4, 0, 0)
now by above solution
equation of line L is
$$\frac{x+4}{0}=\frac{y-1}{1}=\frac{z+1}{-1}$$
let R is an intersection point of line and plane than
$$\frac{x+4}{0}=\frac{y-1}{1}=\frac{z+1}{-1}=k$$
coordinates of R are (–4, k + 1, – k – 1)
this point also lie on plane P, so this will satisfy the equation of plane x + 2y + 3z + 1 = 0
– 4 + 2(k + 1) +3(– k – 1) + 1 = 0
k = – 4
coordinates of R are (–4, –3, 3)
Sol.142.(b)
cos(α+β)cos(α−β)
$\cos^2\alpha - \sin^2\beta$
$\cos^2\alpha - (1 - \cos^2\beta)$
$\cos^2\alpha + \cos^2\beta - 1$
we know that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
so $\cos^2\alpha + \cos^2\beta - 1 = -\cos^2\gamma$
Sol.143.(c)
A(1, 2, −1), B(2, 5, −2) and C(4, 4, −3)
$AB=\sqrt{1+9+1}=\sqrt{11}$
$BC=\sqrt{4+1+1}=\sqrt{6}$
Area of rectangle = length × breadth
$=\sqrt{11}\times\sqrt{6}=\sqrt{66}$
Sol.144.(d)
A(k, 1, −1), B(2k, 0, 2) and C(2 + 2k, k, 1)
Triangle is right angled at B
So line AB and BC are perpendicular
D.R. of AB < k, −1, 3 >
D.R. of BC < 2, k, −1 >
If line AB and BC are perpendicular then
$a_1a_2+b_1b_2+c_1c_2=0$
2k – k – 3 = 0
k = 3
Sol.145.(a)
p = 2q = 3r
let p = 2q = 3r = k
then p = k, q = k/2, r = k/3
p : q : r = 6 : 3 : 2
direction ratio < 6, 3, 2 >
direction cosine $<\frac{6}{7},\frac{3}{7},\frac{2}{7}>$
$$\cos\theta=\frac{3}{7}$$
$$\cos 2\theta=2\cos^2\theta-1=2\left(\frac{3}{7}\right)^2-1=-\frac{31}{49}$$
Sol.146.(b)
direction ratios of line perpendicular to the plane are < 3, 2, 1 >
equation of plane
3x + 2y + z + d = 0
plane passing through the point (1, 1, 1)
3 + 2 + 1 + d = 0
d = – 6
Equation of plane
3x + 2y + z – 6 = 0
Sol.147.(b)
Let line of intersection have D.R. <a, b, c>
This is perpendicular to both given lines
So
6a + 4b – 5c = 0 ……..(i)
a – 2b + 3c = 0 ……..(ii)
$$\frac{a}{12-10}=\frac{b}{-5-18}=\frac{c}{-12-4}$$
$$\frac{a}{2}=\frac{b}{-23}=\frac{c}{-16}$$
D.R. of required line <2, – 23, –16>
Sol.148.(a)
Equation of plane passing through intersection of planes 6x + 4y − 5z = 2 and x − 2y + 3z = 0 is
$$(6x+4y-5z-2)+\lambda(x-2y+3z)=0$$
$$(6+\lambda)x+(4-2\lambda)y+(-5+3\lambda)z-2=0$$
This plane is parallel to the line whose direction ration are <1, 3, 2>
So $a_1a_2 + b_1b_2 + c_1c_2 = 0$
$$(6+\lambda)+(4-2\lambda)3+(-5+3\lambda)2=0$$
$\lambda=-8$
Equation of plane
$$(6x+4y-5z-2)-8(x-2y+3z)=0$$
$$2x-20y+29z+2=0$$
Sol.149.(b)
$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + c = 0$
this sphere is passing through A(1, 0, 0), B(0, 1, 0) and C(0, 0, 1).
1 + 2u + c =0 …….(i)
1 + 2v + c = 0 ……(ii)
1 + 2w + c = 0 ……(iii)
$$u=v=w=\frac{-c-1}{2} \quad \text{….(iv)}$$
Radius
$$r=\sqrt{u^2+v^2+w^2-c}$$
$$r=\sqrt{\left(\frac{c+1}{4}\right)^2+\left(\frac{c+1}{4}\right)^2+\left(\frac{c+1}{4}\right)^2-c}$$
$$r=\sqrt{3\left(\frac{c+1}{4}\right)^2-c}=\sqrt{\frac{3c^2+2c+3}{4}} \quad \text{…(v)}$$
For minimum value of r, $3c^2 + 2c + 3$ should be minimum
Now by concept of maxima and minima
$$\frac{d\left(3c^2+2c+3\right)}{dc}=6c+2=0$$
$$c=-\frac{1}{3}$$
Put this in (v)
$$r=\sqrt{\frac{2}{3}}$$
Sol.150.(a)
Now put value of c in equation (iv) in above solution
$$u=v=w=-\frac{1}{3}$$
Coordinate of centre $(-u,-v,-w)=\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)$
And this point lie on plane x + y + z – 1 = 0 by options.
Sol.151.(b)
Mid point of AC diagonal coincide to mid point of BD diagonal
Let D(x,y,z)
$$\frac{1-1}{2}=\frac{-2+x}{2}\Rightarrow x=2$$
$$\frac{-1+2}{2}=\frac{1+y}{2}\Rightarrow y=0$$
$$\frac{0+7}{2}=\frac{8+z}{2}\Rightarrow z=-1$$
So coordinate of D(2, 0, - 1)
Sol.152.(b)
Direction ratio of BC = <1, 1, - 1>
Direction ratio of CD = <3, - 2, - 8>
Angle between lines
$$\cos\theta=\frac{3-2+8}{\sqrt{1+1+1}\sqrt{9+4+64}}=\frac{9}{\sqrt{3}\sqrt{77}}$$
$$\cos^2\theta=\frac{27}{77}$$